

Crystallography

Matter: Matter is anything that occupies space (volume) and has mass. Matter is made up of extremely small particles (atoms, molecules or ions).

All physical objects are composed of matter, and an easily observed property of matter is its state or phase.

States of Matter:

There are three natural/main states of matter; (1) Solids, (2) liquids, and (3) gases. In addition to these there are two other states of matter, fourth state is the plasma and the fifth state is the man-made Bose-Einstein condensates.

- **Solid:** Atoms /molecules are arranged in a fixed manner and have definite shape and size. In solids, the interatomic (or intermolecular force) is greater than the Kinetic energy of the atoms (or molecules).
- **Liquid & Gases:** Atoms /molecules are not fixed manner and cannot form any shape and size. They gain the shape and size of the container. In liquids & gases, the interatomic (or intermolecular) force is lesser than the Kinetic energy of the atoms (or molecules).
- **Plasma:** It consists of highly charged particles with extremely high kinetic energy.
- **Bose Einstein Condensate (BEC)** (useful in quantum mechanics): A gas of bosons at low densities is cooled to temperature very close to absolute zero.

Solid: Solid is one of the three main states of matter, in which the particles (atoms or molecules or ions) are closely packed together and are not free to move.

On the basis of arrangement of constituent particles solids are divided into,

1. Crystalline solids
2. Amorphous (non-crystalline) solids

Crystalline Solids:

If the atoms or molecules in a solid are arranged in some regular fashion, then the solid is known as crystalline solid.

In a crystalline solid, each atom or molecule is fixed at a definite point in space at a definite distance and in a definite angular orientation to other atoms or molecules surrounding it. The crystalline state of solid is characterized by regular and periodic arrangement of atoms or molecules.

Most of the solids are crystalline in nature and the crystalline solid is further divided into types,

1. Single crystal (Diamond, quartz, mica etc.)
2. Polycrystalline solids (Metals, ceramics etc.)

In single crystal, the periodicity of atoms extends throughout the crystal (Diamond, Quartz).

A polycrystalline material is an aggregate of number of small crystallites with random orientations separated by well-defined boundaries. The smallest crystallites are known as grains and the boundaries are grain boundaries.

It may be noted that although the periodicity of individual crystallites is interrupted at grain boundaries. Yet the polycrystalline form of the material may be more stable compared to the single crystal form. Most of metals and ceramics exhibit polycrystalline structure.

Amorphous Solids:

When the atoms or molecules in a solid are arranged in an irregular fashion, then it is known as amorphous solid.

Amorphous solids are characterized by the complete random arrangement of atoms or molecules. The periodicity if at all present, extends up to a distance of a few atomic diameters only. In other words, these solids exhibit short range order. Amorphous solids are formed when the atoms do not get sufficient time to undergo a periodic arrangement (Glass, plastics, rubber, and wood etc.).

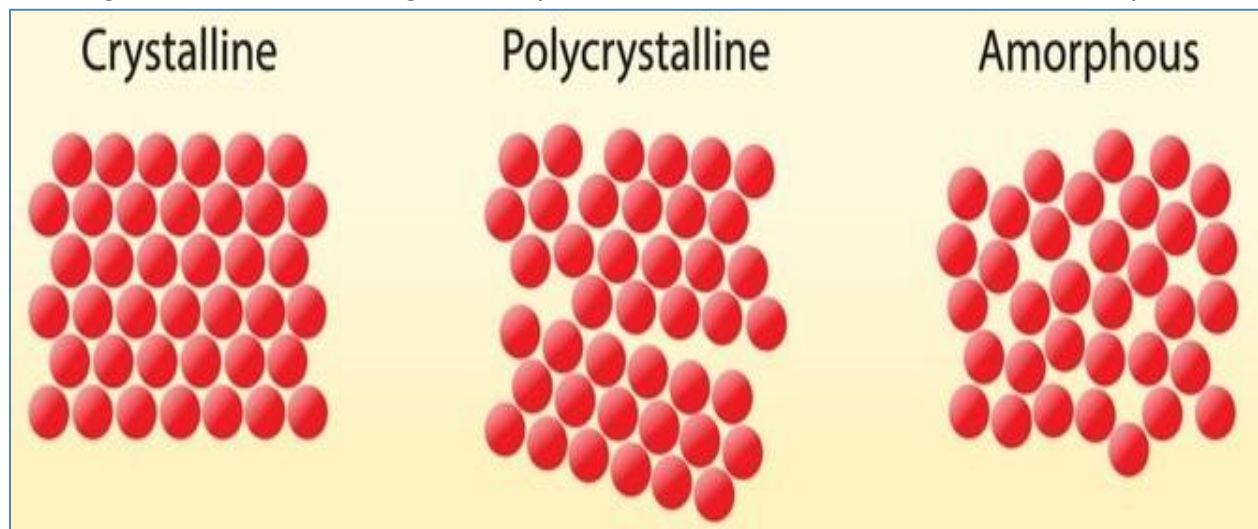


Figure. Crystalline, Poly crystalline and Amorphous solid

Difference between crystalline and amorphous solids

S.No	Crystalline	Amorphous
1	The constituent particles are arranged in a regular fashion	The constituent particles are not arranged in any regular fashion
2	They have definite geometric shapes	They have irregular shapes
3	They are anisotropic (Different properties in different directions)	They are isotropic (Same properties in all directions)
4	They have along range order	They have a short range order
5	They have sharp melting points	They melt over a range of temperature
6	They are ductile in nature	They are brittle in nature
7	They are true solids	They are pseudo solids or supercooled liquids
8	Example- Quartz is crystalline SiO_2	Example-Silica glass is amorphous SiO_2

Crystallography: The branch of science, which deals with the study of geometrical form and physical properties of crystalline solids, is called crystallography.

Crystallography is the experimental science of determining the arrangement of atoms in crystalline solids.

Crystal: A crystal is a three dimensional body, which is made up of regular and periodic arrangement of atoms or molecules in space.

Lattice: The periodic arrangement of points is called lattice.

Space Lattice: It is defined as an array of points in three dimensional space such that the environment about each point is same.

Let us consider the two dimensional array of points as shown in figure 2. It is obvious from the figure that environment about any two points is the same hence it represents a two dimensional space lattice.

If the environment about any points is not the same, then it is not a space lattice.

Lattice points: The points in space lattice are called as lattice points.

Lattice line: The line which joined the lattice points is known as lattice line.

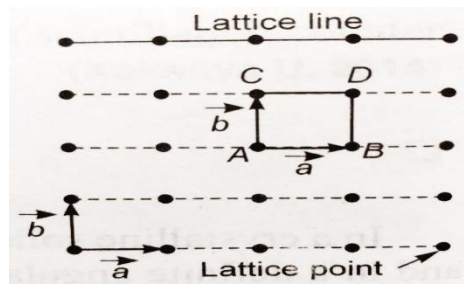


Figure. Space lattice

Basis and crystal structure:

A set of one or more atoms (unit assembly of atoms) about a lattice point is called a basis.

The space lattice together with the basis gives the actual crystal structure i.e., Crystal structure is formed when a basis of atoms is attached identically to each lattice point.

When the basis is repeated with correct periodicity in all directions it gives the actual crystal structure. The crystal structure is real, while the lattice is imaginary. Mathematically it is expressed as

Space lattice + basis -----> Crystal structure

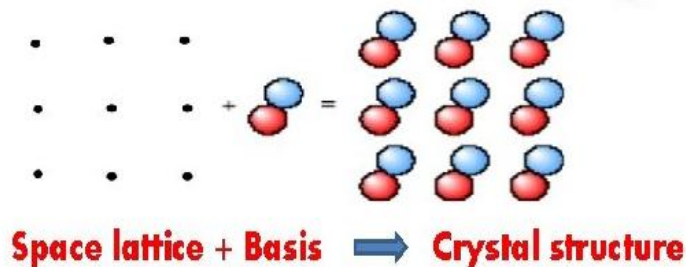


Figure. Crystal structure

The number of atoms in a basis varies from one to several thousands. In crystalline solids like Cu and Na the basis is Single atom, In NaCl & CsCl the basis is diatomic, whereas in crystals, like CaF_2 the basis is tri-atomic. Complex basis are found in biological materials.

Translation vectors or Primitive vectors:

Lattice points in space can be located with the help of a coordinate system. For example, consider a coordinate system with its origin at lattice point A as shown in figure 2. From the figure, A, B, and C are nearest non-collinear points. Let $AB = a$ and $AC = b$. Here a and b are known as coordinate vectors. The position vector R of any lattice point can be expressed as $R = n_1a + n_2b$

where n_1 and n_2 are integers, whose values depend upon the location of

lattice point with respect to origin. Now the vectors a and b are also termed as basis or translation vector. The arrangement of lattice points in three dimensions is expressed by $R = n_1a + n_2b + n_3c$; In this equation, a , b , and c are called primitives.

Unit cell and lattice parameters:

The unit cell is the smallest block or geometrical figure the repetition of which gives the actual crystal structure and represents fully all the characteristics of the crystal. (Or)

Unit cell may be defined as the volume of a solid from which entire crystal may be constructed by translational repetition in three dimensions.

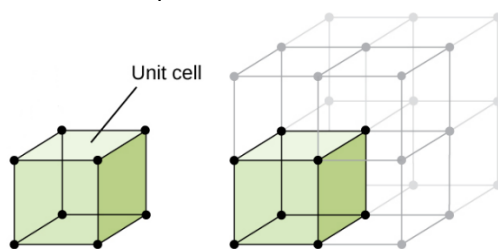


Figure. Unit Cell

Primitive cell

In a primitive cell, all the lattice points lie at its corners. Therefore, the **effective number of lattice points in a primitive cell is one.**

Non-primitive cell

In a non-primitive cell, lattice points lie at the corners as well as at other locations, both inside and on the surface of the cell. Therefore, the effective number of lattice points in a non-primitive cell is greater than one.

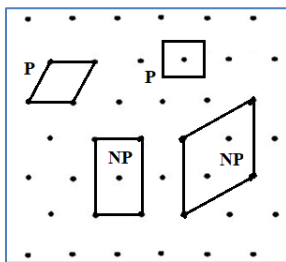


Figure. Primitive(P) and non-primitive (NP) Unit cells of 2-D lattice

Lattice Parameters of Unit Cell:

Axial lengths or intercepts: The distance between two neighboring points is called as the edges of the unit cell. The lengths OE , OC and OA along the three axes OX , OY , and OZ are called as axial lengths or intercepts. The axial lengths along the three axes are represented by vectors a , b , and c respectively.

Interfacial angles: The angles between the above three intercepts are called interfacial angles (γ , α , and β). Or The angles between three crystallographic axes are known as interfacial angles or inter-axial angles.

Both intercepts (a , b , and c) and interfacial angles (γ , α , and β) together are known as lattice parameters of unit cell (or crystal). These parameters determine the actual space and size of unit cell as shown in figure 6.

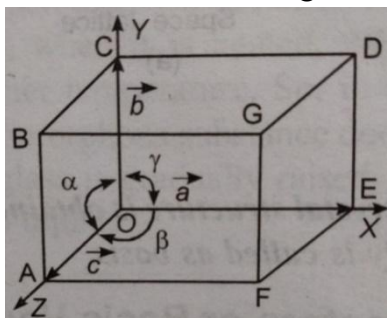


Figure. Lattice parameters (Crystallographic axis and angles) of Unit Cell

❖ **Expression for Lattice Constant of a Cubic crystal (a):-**

Let us consider a cubic unit cell in which 'a' is the edge length or primitive of a cubic unit cell and 'ρ' is the density of the crystal.

Density $\rho = (\text{total mass of the unit cell}) / (\text{volume of unit cell})$

The mass of the unit cell = ρa^3 ----- (i);

Where, $a^3 \rightarrow$ Volume of the cubic unit cell.

Let 'M' be the molecular weight and ' N_A ' is the Avogadro number (i.e. no. of molecules per Kg mole of substance)

Then the mass of the each molecule = M / N_A ; If each unit cell contains 'n' molecules, then mass of the unit cell = $n M / N_A$ ----- (ii)

From equation (i) and (ii)

$$\rho a^3 = n M / N_A$$

$$a^3 = n M / \rho N_A$$

Lattice Constant for Cubic Lattice, **$a = (n M / \rho N_A)^{1/3}$** .

Where, $n \rightarrow$ no. of molecules, $N_A \rightarrow$ Avogadro number, $M \rightarrow$ Molecular weight, $\rho \rightarrow$ Density of unit cell, $a \rightarrow$ Lattice constant / parameter of the unit cell/ crystal.

Crystal systems:

The vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ (axial lengths vectors) may or may not be equal. Also the interfacial angles α , β and γ may or may not be right angles.

Based on these conditions, there are seven crystal systems. All these crystal systems are divided to 7 groups or basic systems. The seven basic crystal systems are distinguished from one another by the angles between the three axes and the intercepts of the faces along them.

On the basis of the length and directions of the axes of the symmetry all the crystals may be classified in to following seven systems.

The basic crystal systems are 1) Cubic, 2) Tetragonal, 3) Orthorhombic, 4) Rhombohedral or Trigonal, 5) Hexagonal, 6) Monoclinic, and 7) Triclinic.

1. **Cubic crystal:** In this crystal, all the three axial lengths of the unit cell are equal and the three axes are mutually perpendicular to each other. I.e., $a=b=c$ & $\alpha=\beta=\gamma=90^\circ$.
Examples: NaCl, CaF₂, Diamond, Cu, etc.
2. **Tetragonal crystal:** In this crystal, two axial lengths of the unit cell are equal while the third one is longer and the three axes are mutually perpendicular to each other. I.e., $a=b \neq c$ & $\alpha=\beta=\gamma=90^\circ$.
Examples: TiO₂, SnO₂, NiSO₄, etc.
3. **Orthorhombic crystal:** In this crystal, all the three axial lengths of the unit cell are different (unequal) and the three axes are mutually perpendicular to each other. I.e., $a \neq b \neq c$ & $\alpha=\beta=\gamma=90^\circ$.
Examples: Ga, BaSO₄, KNO₃, etc.
4. **Rhombohedral or Trigonal crystal:** In this crystal, all the three axial lengths of the unit cell are equal and all the three axes are equally inclined to each other at an angle other than 90° . I.e., $a=b=c$ & $\alpha=\beta=\gamma \neq 90^\circ$. Examples: As, Sb, Bi, etc.
5. **Hexagonal crystal:** In this crystal, two axial lengths of the unit cell are equal while the third one is longer and the two axes of the unit cell are equal and perpendicular to each other and the third one is at 120° . I.e., $a=b \neq c$ & $\alpha=\beta=90^\circ, \gamma=120^\circ$.
Examples: SiO₂, AgI, Mg, Zn, Cd, etc.
6. **Monoclinic crystal:** In this crystal, all the three axial lengths of the unit cell are different (unequal) and two axes are right angles and third one is obliquely inclined. I.e., $a \neq b \neq c$ & $\alpha=\beta=90^\circ \neq \gamma$.
Examples: FeSO₄, CaSO₄.2H₂O, etc.
7. **Triclinic crystal:** In this crystal, all the three axial lengths of the unit cell are

different (unequal) and also the three axes are not right angles and not equal to each other. I.e., $a \neq b \neq c$ & $\alpha \neq \beta \neq \gamma \neq 90^\circ$.

Examples: CuSO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$, etc.

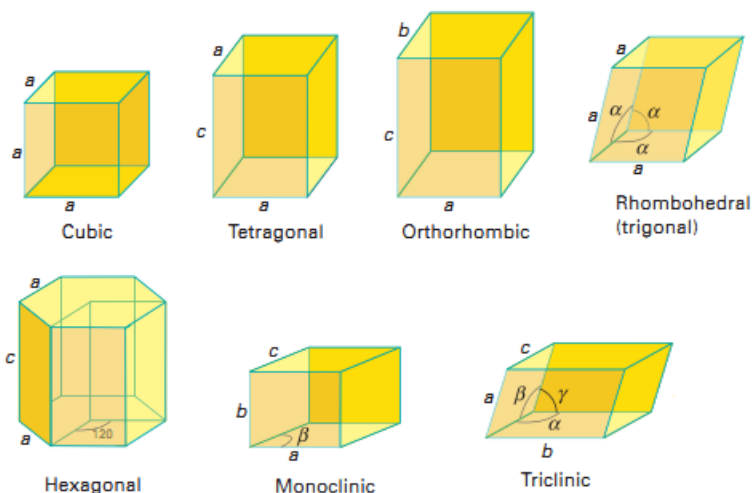


Fig. Crystal Systems

Seven crystal systems:

Sl. No.	Name of the Crystal system	Lattice parameters		Examples
		Primitives/ Axial lengths	Interfacial Angles	
1.	Cubic	$a=b=c$	$\alpha=\beta=\gamma=90^\circ$	NaCl , CaF_2 , Diamond, Cu, etc.
2.	Tetragonal	$a=b \neq c$	$\alpha=\beta=\gamma=90^\circ$	NiSO_4 , SnO_2 , TiO_2 , etc.
3.	Orthorhombic	$a \neq b \neq c$	$\alpha=\beta=\gamma=90^\circ$	BaSO_4 , KNO_3 , Ga, etc.
4.	Rhombohedral(or) Trigonal	$a=b=c$	$\alpha=\beta=\gamma \neq 90^\circ$	As, Sb, Bi, etc.
5.	Hexagonal	$a=b \neq c$	$\alpha=\beta=90^\circ, \gamma=120^\circ$	SiO_2 , Mg, Zn, Cd, etc.
6.	Monoclinic	$a \neq b \neq c$	$\alpha=\beta=90^\circ \neq \gamma$	FeSO_4 , $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$, etc.
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	CuSO_4 , $\text{K}_2\text{Cr}_2\text{O}_7$ etc.

P: Primitive, **I:** Body centered **F:** Face centered **C:** Base (end) centered.

Bravais Lattices:

In the year of 1948, Bravais showed that there are only fourteen independent ways of arranging the points in three-dimensional space such that each arrangement confirms to the definition of space lattice. So, there are 14 lattices are sufficient to describe all the crystals.

Definition: There are fourteen different ways of arranging the points in three-dimensional space such that each point has identical surrounding. These 14 space lattices are known as Bravais lattices.

- ❖ **Simple or Primitive (P):** If the atoms are placed only at the corner of the unit cell (cube) then it is called primitive and is denoted by 'P'.
- ❖ **Body centered (I):** If the atoms are placed at the corners of the unit cell (cube) and the center of the unit cell (where body diagonals intersect) then it is said to be body centered and is denoted by 'I'.
- ❖ **Face Centered (F):** If the atoms are placed at the corners of the unit cell (cube) and the center of their faces then it is said to be Face centered and is denoted by 'F'.
- ❖ **Base centered (C):** If the atoms are placed at the corners of the unit cell (cube) and the center of the two bases of the unit cell (cube) then it is called base (end) centered and is denoted by 'C'.

The diagrams of the unit cells are drawn below.

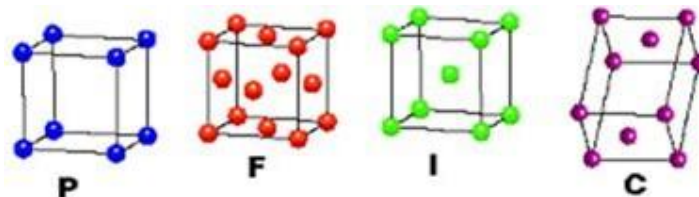


Fig. Types of unit cells

1. Cubic crystal structure: (P, I, F)

In this crystal structure, there exists three (3) lattices and they are: Simple or Primitive (P), Body centered (I), and Face centered (F).

2. Tetragonal crystal structure: (P, I)

In this crystal structure, there are only two (2) lattices exists and they are: Simple or Primitive (P), and Body centered (I).

3. Orthorhombic crystal structure: (P, I, F, C)

In this crystal structure, there are only four (4) lattices exists and they are: Simple or Primitive cubic (P), Body centered cubic (I), Face centered cubic (F) and Base centered (C).

4. Rhombohedral or Trigonal crystal structure: (P)

In this crystal structure, there is only one (1) lattice exists and that is Simple or Primitive (P).

5. Hexagonal crystal structure:(P)

In this crystal structure, there is only one (1) lattice exists and that is Simple or Primitive (P).

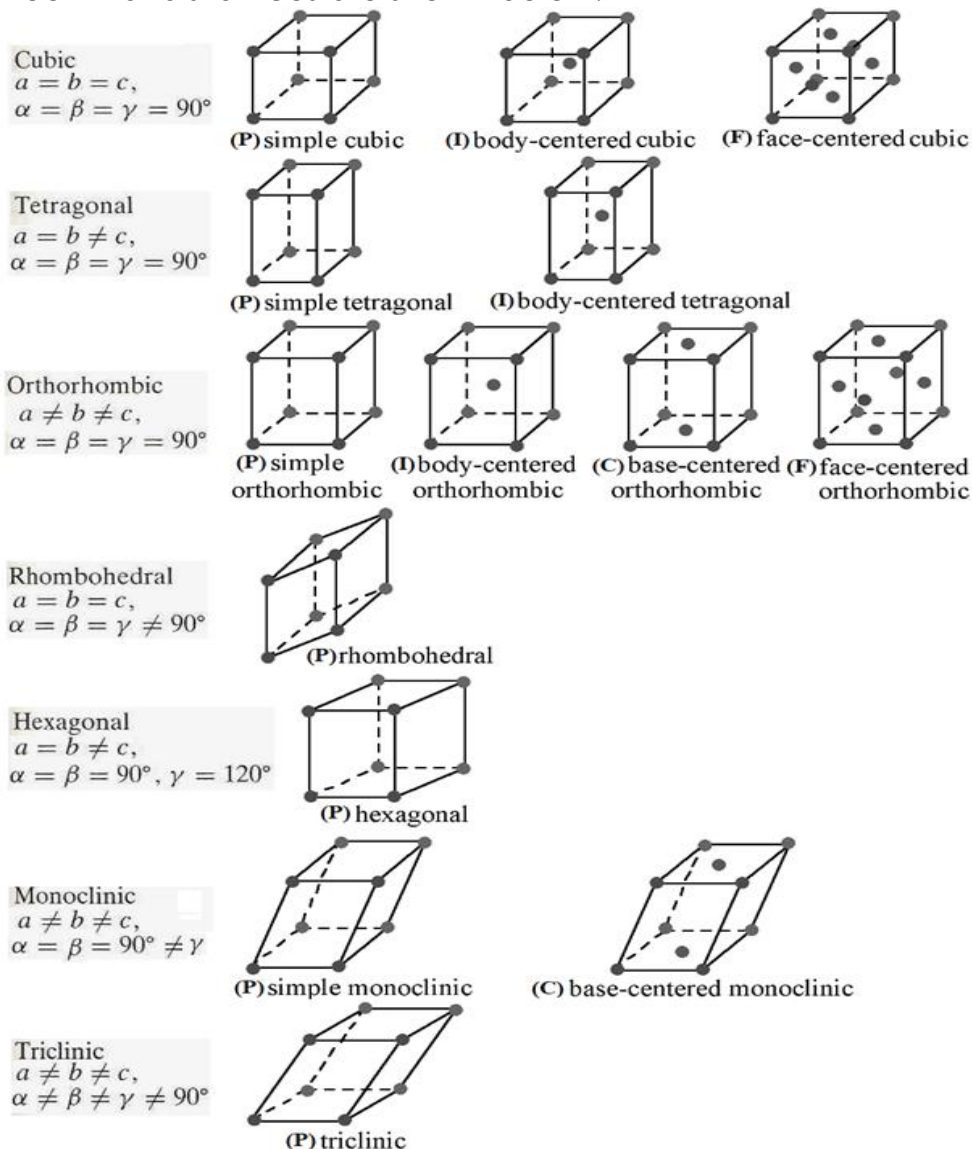
6. Monoclinic crystal structure:(P, I)

In this crystal structure, there are only two (2) lattices exists and they are: Simple or Primitive (P), and Body centered (I).

7. Triclinic crystal structure:(P)

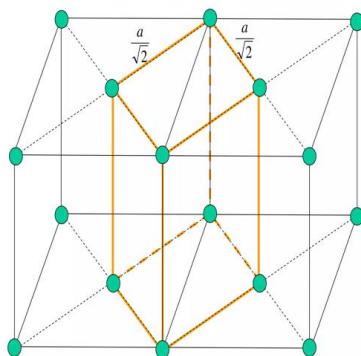
In this crystal structure, there is only one (1) lattice exists and that is Simple or Primitive (P).

The fourteen Bravais lattices are shown below:



Why End (Base) centered cubic not in the Bravais list? Why can't a body centered tetragonal equivalent to Cubic F?

Based on the unit cell approach, from careful observation, we can find a small tetragonal P unit cell, so, it was concluded that the tetragonal P was a better description for cubic C.



End-centred cubic = Simple Tetragonal

On the other hand, when we apply the same approach to cubic F, we can select a smaller unit cell i.e. a tetragonal I (body centered tetragonal) but why not the same argument tetragonal I is a better description for cubic F, if so, we can reduce the no. of Bravais lattices to 13. This difficulty leads to look into the symmetry-based approach.

❖ Symmetry:

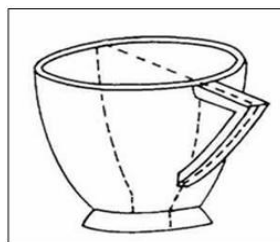
If an object is brought into self-coincidence after some operation it said to possess symmetry with respect to that operation.

(Or) Symmetry is a property of an object to stay unchanged (invariant) after certain type of movement.

A Symmetry operation: An operation that takes the crystal into itself is called a symmetry operation. i.e. when a symmetry operation is performed on lattice, it leaves the lattice invariant.



Rotation by 60 degrees



Reflection in the plane through the middle

According to the type of movement there are different types of symmetry operations. **1. Point symmetry, 2. Space symmetry**

Point symmetry: AT LEAST one point is fixed during the movement of the object
(Non-translation symmetry)

- Rotation
- Reflection
- Inversion (Centre of symmetry or inversion center)
- Improper rotation (Combination of above) i.e., Rotoreflexion/ rotoinversion.

Rotational Symmetry:

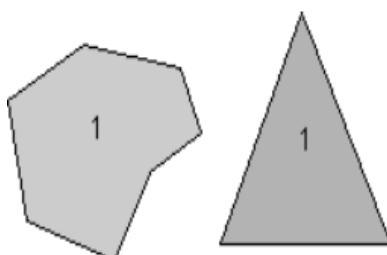
If the crystal is invariant under rotation about an axis through an angle $\phi = 2\pi/n$, then the axis of rotation is said to have n-fold symmetry ($n = 2\pi/\phi$).

If an object can be rotated about an axis and repeats itself every 90° of rotation then it is said to have an axis of 4-fold rotational symmetry. The axis along which the rotation is performed is an element of symmetry referred to as a rotation axis.



The following types of rotational symmetry axes are possible in crystals.

1-Fold Rotation Axis: An object that requires rotation of a full 360° in order to restore it to its original appearance has no rotational symmetry. Since it repeats itself 1 time every 360° it is said to have a 1-fold axis of rotational symmetry.



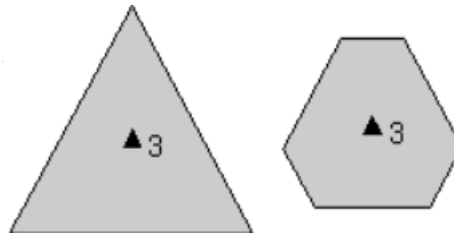
2-fold Rotation Axis: If an object appears identical after a rotation of 180° , that is twice in a 360° rotation, then it is said to have a 2-fold rotation axis ($360/180 = 2$).

(Note that in these examples the axes we are referring to are imaginary lines that extend toward you perpendicular to the page or blackboard)

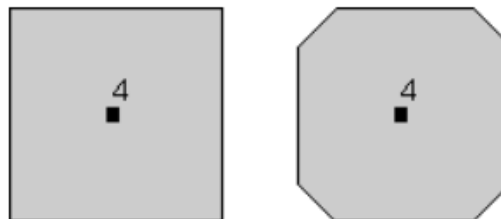
A filled small oval shape represents the point where the 2-fold rotation axis intersects the page. This symbolism will be used for a 2-fold rotation axis.



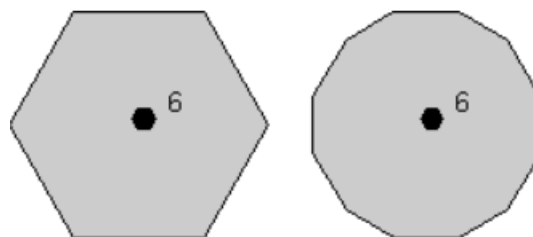
3-Fold Rotation Axis: Objects that repeat themselves upon rotation of 120° are said to have a 3-fold axis of rotational symmetry ($360/120=3$), and they will repeat 3 times in a 360° rotation. A filled triangle is used to symbolize the location of 3-fold rotation axis.



4-Fold Rotation Axis: If an object repeats itself after 90° of rotation, it will repeat 4 times in a 360° rotation, as illustrated previously. A filled square is used to symbolize the location of 4-fold axis of rotational symmetry.



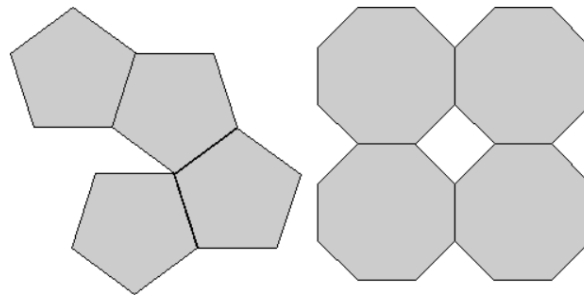
6-Fold Rotation Axis: If rotation of 60° about an axis causes the object to repeat itself, then it has 6-fold axis of rotational symmetry ($360/60=6$). A filled hexagon is used as the symbol for a 6-fold rotation axis.



The rotational symmetries allowed are only 1, 2, 3, 4 and 6-fold.

***Although objects themselves may appear to have 5-fold, 7-fold, 8-fold, or higher-fold rotation axes, these are not possible in crystals. The reason is that the external shape of a crystal is based on a geometric arrangement of atoms. Note that if we try to combine objects with 5-fold and 8-fold apparent symmetry, that we cannot combine them in such a way that they completely fill space, as

illustrated below.

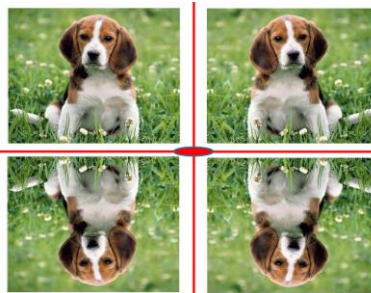


Reflection symmetry (Mirror Symmetry):

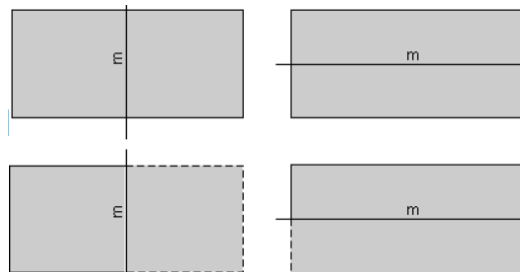
If the reflection in the mirror reproduces the other half of the object, then the object is said to have mirror symmetry.

A mirror symmetry operation is an imaginary operation that can be performed to reproduce an object. The operation is done by imagining that you cut the object in half, then place a mirror next to one of the halves of the object along the cut.

The plane of the mirror is an element of symmetry referred to as a mirror plane, and is symbolized with the letter "m".



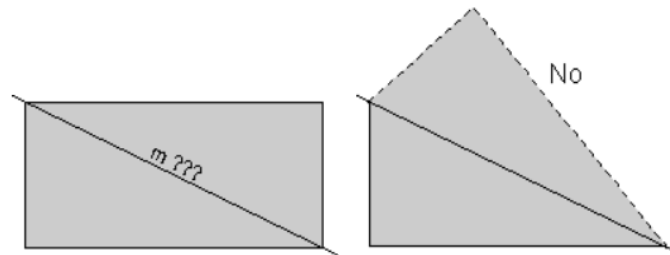
The dashed parts of the rectangles below show the part the rectangles that would be seen as a reflection in the mirror.



The rectangles shown above have two planes of mirror symmetry. Three dimensional and more complex objects could have more. For example, the hexagon shown above, not only has a 6-fold rotation axis, but has 6 mirror planes.

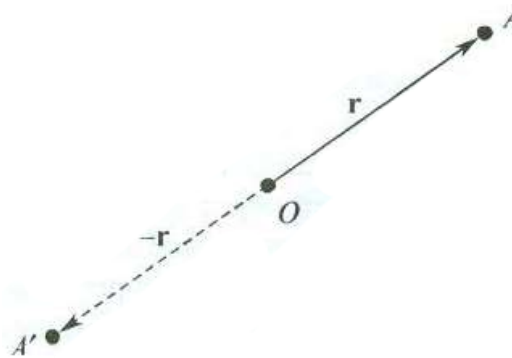
***Note that a rectangle does not have mirror symmetry along the diagonal lines. If we cut the rectangle along a diagonal such as that labeled "m ???", as shown

in the upper diagram, reflected the lower half in the mirror, then we would see what is shown by the dashed lines in lower diagram. Since this does not reproduce the original rectangle, the line "m???" does not represent a mirror plane.



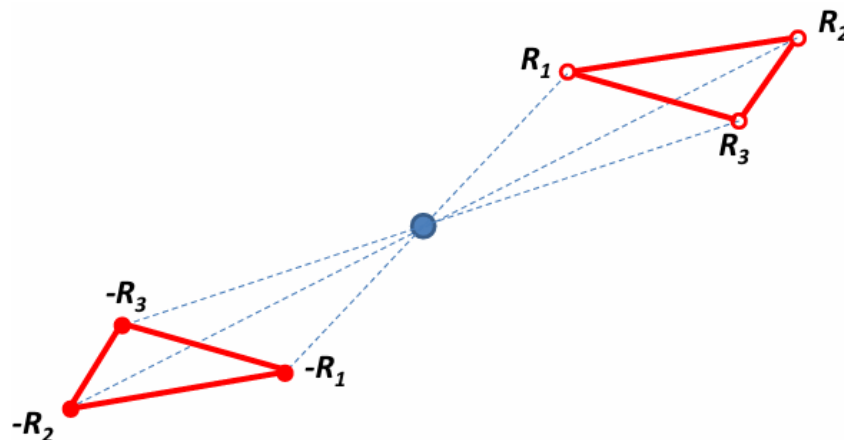
Inversion Symmetry (Centre of symmetry or inversion center):

If to every lattice point at r there is another lattice point at position vector $-r$, then we say that the lattice has inversion symmetry. The origin is said to be the centre of inversion symmetry, symbolically represented by $\bar{1}$.



The lines each have lengths that are equidistant from the original points. When the ends of the lines are connected, the original object is reproduced inverted from its original appearance.

In the diagram shown here, only a few such lines are drawn for the small triangular face. The right-hand diagram shows the object without the imaginary lines that reproduced the object.



Improper Rotation:

Rotoreflexion: If a rotation and a reflection is combined to form a single hybrid operation, the resulting operation is called rotoreflexion.

It is a combined geometric symmetry operation involving a rotation around an axis followed by a reflection in a plane perpendicular to that axis.

It is represented by a tilde over the corresponding rotation, e.g. $\tilde{1}, \tilde{2}$ etc.

Rotoinversion: If rotation is combined with inversion, it produces rotoinversion. It is represented by placing a bar over the numerical symbol corresponding to rotation axis, e.g. $\bar{1}, \bar{2}$, etc.

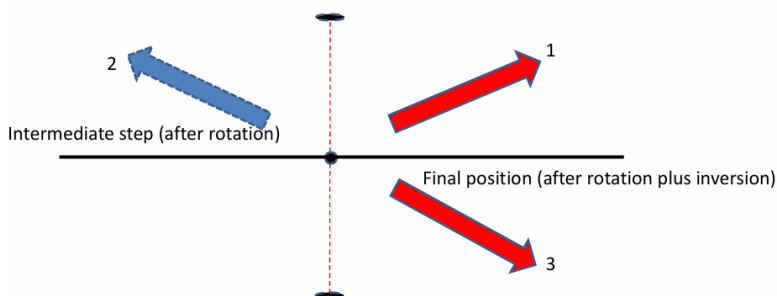
Such an axis has the symbol $\bar{1}$ over bar ($\bar{1}$), as shown in the right-hand diagram above. Note that crystals that have a center of symmetry will exhibit the property that if you place it on a table there will be a face on the top of the crystal that will be parallel to the surface of the table and identical to the face resting on the table. Objects that have rotoinversion symmetry have an element of symmetry called a rotoinversion axis.

A 1-fold rotoinversion axis is the same as a center of symmetry.

Other possible rotoinversion are as follows:

2-fold Rotoinversion: The operation of 2-fold rotoinversion involves first rotating the object by 180° then inverting it through an inversion center.

This operation is equivalent to having a mirror plane perpendicular to the 2-fold rotoinversion axis and would be pronounced as " $\bar{2}$ ". But, since this the equivalent of a mirror plane, m , the bar 2 is rarely used.



Designation of the rotoinversion axes:

$\bar{2} \equiv m$	
$\bar{3}$	
$\bar{4}$	
$\bar{6}$	

Point Groups:

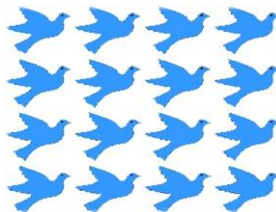
- ✚ Groups formed by combination of rotation, reflection and inversion symmetry operations about a fixed point forms a point group.
- ✚ In this symmetry operations at least one point remains at rest.
- ✚ In 3 dimensions addition of inversion leads to a total of 32 point groups.

Space symmetry: no points are fixed during the movement of the object.

- Translation (lattice translation)
- Combination of translation and any kind of point symmetry operations.

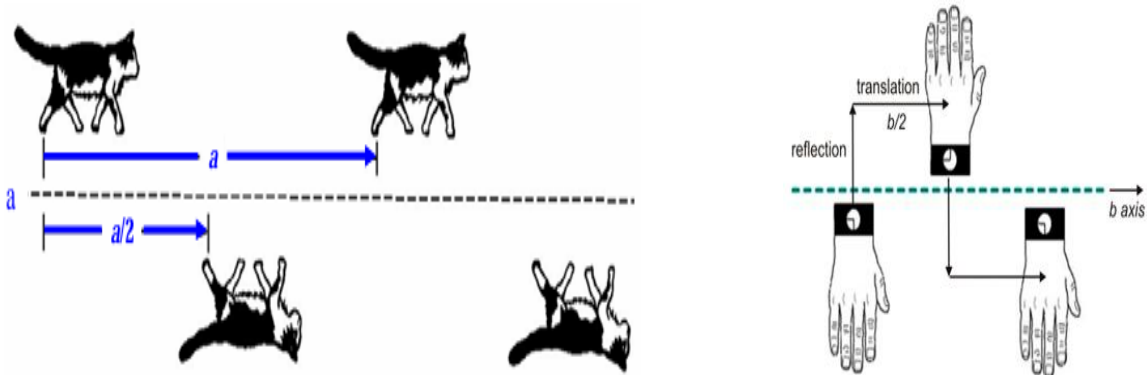
Translation symmetry:

Translation symmetry means that the environment about a point in the lattice remains unchanged when translated by lattice translation vector $T = n_1a + n_2b + n_3c$. Translation symmetry is the symmetry in which an object moves from one position to another, with the same orientation in the forward and backward motion.



Glide plane (Glide reflection):

A glide plane represents a translation followed by a reflection in a mirror plane.



Glide planes are symbolized as: a, b, c, n, & d.

a: → glide parallel to the x direction and a translation of $a/2$

b: → glide parallel to the y direction and a translation of $b/2$

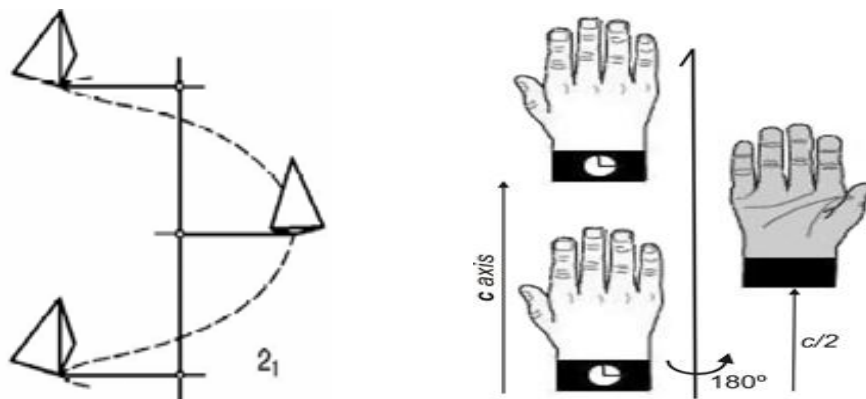
c: → glide parallel to the z direction and a translation of $c/2$

n: → glide along the face diagonal and a translation of $1/2$ of the face diagonal

d: → glide along the body diagonal and a translation of $1/4$ of the body diagonal.

Screw axis (Screw rotations):

The screw axis represents a translation followed by rotation. A screw axis is represented as N_n where N is the order of rotation and the translation distance is n/N . For example: 2_1 represents a 2 fold rotation followed by a translation of $1/2$ parallel to the axis of rotation.



The seven crystal systems based on symmetry:

Sl.No.	Symmetry	Crystal System
1.	1-fold symmetry	Cubic
2.	A single 2-fold 	Tetragonal
3.	A single 3-fold 	Orthorhombic
4.	A single 6-fold 	Rhombohedral or Trigonal
5.	Three 2-fold 3 	Hexagonal
6.	A single 4-fold 	Monoclinic
7.	Four 3-fold 4 	Triclinic

Space Groups:

- ✚ The group of all symmetry operations (i.e. translations as well as point symmetry) of a crystal structure is called a space group.
- ✚ Space groups are combinations of translational symmetry of unit cell and symmetry operations such as rotation, reflection, roto-reflection, roto-inversion, screw translation and glide reflection symmetry operations.
- ✚ It determines the symmetry of crystal structure as a whole. There are 230 distinct space groups in three dimensions.

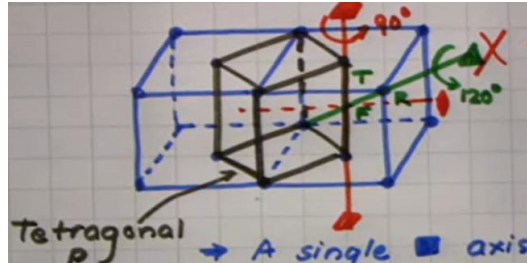
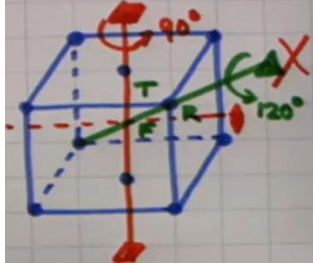
triclinic			tetragonal			cubic							
1	P_1	$\bar{1}$	37	$Ccc2$	75	$P4$	115	$P-4m2$	154	$P3_221$	193	$P6_3/mcm$	
2	$P-1$	$\bar{1}1$	38	$Am m2$	76	$P4_1$	116	$P-4c2$	155	$R32$	194	$P6_3/mmc$	
monoclinic			39	$Abm2$	77	$P4_2$	117	$P-4b2$	156	$P3m1$	cubic		
3	$P2$	2	40	$Ama2$	78	$P4_3$	118	$P-4n2$	157	$P31m$	195	$P23$	
4	$P2_1$	2_1	41	$Ab a2$	79	$I4$	119	$I-4m2$	158	$P3c1$	196	$F23$	
5	$C2$	2	42	$Fmm2$	80	$I4_1$	120	$I-4c2$	159	$P31c$	197	$I23$	
6	Pm	m	43	$Fdd2$	81	$P-4$	121	$I-42m$	160	$R3m$	198	$P2_13$	
7	Pc	m	44	$Im m2$	82	$I-4$	122	$I-42d$	161	$R3c$	199	$I2_13$	
8	Cm	$2/m$	45	$Iba2$	83	$P4/m$	123	$P4/m m m$	162	$P-31m$	200	$Pm3$	
9	Cc	$2/m$	46	$Im a2$	84	$P4_2/m$	124	$P4/m c c$	163	$P-31c$	201	$Pn3$	
10	$P2/m$	$2/m$	47	$Pmm m$	85	$P4/n$	125	$P4/n b m$	164	$P-3m1$	202	$Fm3$	
11	$P2_1/m$	$2_1/m$	48	$Pnn n$	86	$P4_2/n$	126	$P4/n n c$	165	$P-3c1$	203	$Fd3$	
12	$C2/m$	$2/m$	49	$Pcc m$	87	$I4/m$	127	$P4/m b m$	166	$R-3m$	204	$Im3$	
13	$P2/c$	$2/c$	50	$Pban$	88	$I4_1/a$	128	$P4/m n c$	167	$R-3c$	205	$Pa3$	
14	$P2_1/c$	$2_1/c$	51	$Pmma$	89	$P422$	129	$P4/n m m$	hexagonal			206	$Ia3$
15	$C2/c$	$2/c$	52	$Pnna$	90	$P42,2$	130	$P4/n c c$	168	$P6$	207	$P432$	
orthorhombic			53	$Pmna$	91	$P4_2,2$	131	$P4_2/m m c$	169	$P6_1$	208	$P4_232$	
16	$P222$	222	54	$Pcca$	92	$P4_2,2$	132	$P4_2/m c m$	170	$P6_5$	209	$F432$	
17	$P222_1$	222_1	55	$Pbam$	93	$P4_2,2$	133	$P4_2/n b c$	171	$P6_2$	210	$F4_132$	
18	$P2_12_12_1$	$2_12_12_1$	56	$Pccn$	94	$P4_2,2$	134	$P4_2/n n m$	172	$P6_4$	211	$I432$	
19	$P2_1,2,2_1$	$2_12_12_1$	57	$Pbcm$	95	$P4_2,2$	135	$P4_2/m n m$	173	$P6_3$	212	$P4_332$	
20	$C222_1$	222_1	58	$Pnnm$	96	$P4_3,2$	136	$P4_2/n m c$	174	$P-6$	213	$P4_132$	
21	$C222$	222	59	$Pmmn$	97	$I422$	137	$P4_2/n c m$	175	$P6/m$	214	$I4_132$	
22	$F222$	222	60	$Pbcn$	98	$I4_1,2$	138	$I4/m m m$	176	$P6_3/m$	215	$P-43m$	
23	$I222$	222	61	$Pbca$	99	$P4 m m$	139	$I4/m c m$	177	$P622$	216	$F-43m$	
24	$I2_12_12_1$	$2_12_12_1$	62	$Pnma$	100	$P4 b m$	140	$I4_1/a m d$	178	$P6_122$	217	$I-43m$	
25	$Pmm2$	222	63	$Cmcm$	101	$P4_2 c m$	141	$I4_1/a c d$	179	$P6_522$	218	$P-43n$	
26	$Pmc2_1$	222	64	$Cmca$	102	$P4_2 n m$	142		180	$P6_222$	219	$F-43c$	
27	$Pcc2$	222	65	$Cmm m$	103	$P4 c c$	trigonal			181	$P6_22$	220	$I-43d$
28	$Pma2$	222	66	$Cccm$	104	$P4 n c$	143	$P3$	182	$P6_322$	221	$Pm3m$	
29	$Pca2_1$	222	67	$Cmma$	105	$P4_2 m c$	144	$P3_1$	183	$P6 m m$	222	$Pn3n$	
30	$Pnc2$	222	68	$Ccca$	106	$P4_2 b c$	145	$P3_2$	184	$P6 c c$	223	$Pm3n$	
31	$Pmn2_1$	222	69	$Fmm m$	107	$I4 m m$	146	$R3$	185	$P6_3 c m$	224	$Pn3m$	
32	$Pba2$	222	70	$Fddd$	108	$I4 c m$	147	$P-3$	186	$P6_3 m c$	225	$Fm3m$	
33	$Pna2_1$	222	71	$Im m m$	109	$I4_1 m d$	148	$R-3$	187	$P-6m2$	226	$Fm3c$	
34	$Pnn2$	222	72	$Ibam$	110	$I4_1 c d$	149	$P312$	188	$P-6c2$	227	$Fd3m$	
35	$Cmm2$	222	73	$Ibca$	111	$P-42m$	150	$P321$	189	$P-62m$	228	$Fd3c$	
36	$Cmc2_1$	222	74	$Imma$	112	$P-42c$	151	$P3_12$	190	$P-62c$	229	$Im3m$	
					113	$P-4_21m$	152	$P3_21$	191	$P6/m m m$	230	$Ia3d$	
					114	$P-4_21c$	153	$P3_212$	192	$P6/m c c$			

Table: 14 Bravais Lattices with 7 crystal systems:

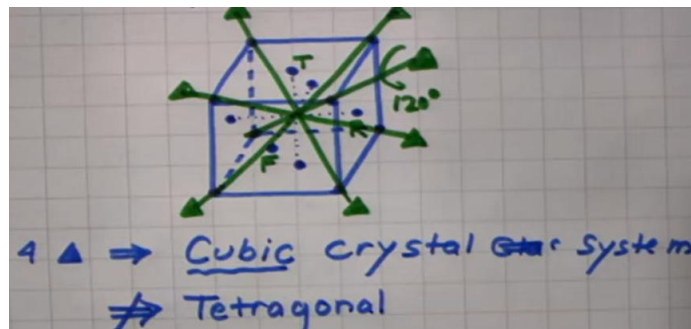
Crystal System	Lattice Parameters	Types of Bravais lattices	No. of Bravais lattices	Examples
1. Cubic	$a=b=c$ & $\alpha=\beta=\gamma=90^\circ$	P, I, F	3	NaCl, CaF ₂ , Diamond, etc.
2. Tetragonal	$a=b \neq c$ & $\alpha=\beta=\gamma=90^\circ$	P, I	2	NiSO ₄ , SnO ₂ , TiO ₂ , etc.
3. Orthorhombic	$a \neq b \neq c$ & $\alpha=\beta=\gamma=90^\circ$	P, I, F, C	4	BaSO ₄ , KNO ₃ , Ga, etc.
4. Rhombohedral or Trigonal	$a=b=c$ & $\alpha=\beta=\gamma \neq 90^\circ$	P	1	As, Sb, Bi, etc.
5. Hexagonal	$a=b \neq c$ & $\alpha=\beta=90^\circ, \gamma=120^\circ$	P	1	SiO ₂ , Mg, Zn, Cd, etc.
6. Monoclinic	$a \neq b \neq c$ & $\alpha=\beta=90^\circ \neq \gamma$	P, C	2	FeSO ₄ , CaSO ₄ .2H ₂ O, etc.
7. Triclinic	$a \neq b \neq c$ & $\alpha \neq \beta \neq \gamma \neq 90^\circ$	P	1	CuSO ₄ , K ₂ Cr ₂ O ₇ etc.

A symmetry-based approach to Bravais Lattices:

Based on the symmetry approach, in Cubic C (see the figure below): 3-fold axis is not possible; 2-fold axis not possible; only 4-fold axis is possible i.e. in tetragonal system, we will be having 4-fold axis, so Cubic C is not possible, and it is a tetragonal P.



Let us consider a cubic F (see the figure below), start looking at body diagonal and rotate by 120° then top face will come at the place of front face and the front face will move to the right face, so each time the cube and lattice will come into self-coincidence by this 3-fold rotation symmetry. So, like this we will have all four 3-fold ($4 \blacktriangle$) axis. Hence, it is a cubic and it is not possible to replace it with Tetragonal I.



- **Wyckoff positions:** (Tell where the atoms in a crystal we can find)
- A Wyckoff position is a set of points in a unit cell that are equivalent under all symmetry operations of a space group and have the same site symmetry.
- They tell us where atoms can sit in a crystal structure while remaining consistent with the space-group symmetry.
- A Wyckoff position is designated by the Wyckoff label, for instance, 8a. The '8' is the multiplicity, indicating how many points belong to an orbit of the Wyckoff position are included in a unit cell.
- The 'a' is an alphabetical label, i.e. small Roman letter, (a, b, c, ...),

representing different Wyckoff positions.

- The maximum number of different Wyckoff positions in the 230 space groups is 27 (in Group (47) Pmmm).
- The various possible sets of Wyckoff positions for all the space groups are given in the International Tables for Crystallography [Refer: International Tables for Crystallography. Volume A: Space-group symmetry (Th. Hahn, ed.). Kluwer, Dordrecht, (editions of 1983, 1987, 1992, 1995, 2002, 2005)].

Each Wyckoff position is uniquely defined by: Multiplicity, Wyckoff letter, Site symmetry, and Coordinates.

1. Multiplicity

number of symmetry-equivalent positions generated by space-group operations.

Example: 2a → 2 equivalent positions; 6c → 6 equivalent positions

Multiplicity depends on: Space group, Position symmetry.

2. Wyckoff letters:

Letters are assigned alphabetically: a, b, c, d, ...

a → highest site symmetry

Later letters → lower symmetry, more positional freedom

3. Site symmetry:

Site symmetry is the subgroup of symmetry operations that leave that point unchanged.

Notation examples: **m, 2, 3m, -1**

Higher site symmetry → fewer degrees of freedom.

4. Coordinates in Wyckoff positions:

Wyckoff positions specify atomic coordinates:

(a) Fixed positions: All coordinates fixed by symmetry. Example: 4a: (0,0,0)

(b) Partially free: Some coordinates are variable. Example: 6c: (0,0,z)

(c) General positions: All coordinates free. Example: 12i: (x,y,z)

Some Examples:

1. LiCoO_2 (Layered oxide cathode)

Crystal system: Trigonal (rhombohedral, hexagonal setting)

Space group: R-3m (No. 166)

Atom	Wyckoff position	Multiplicity	Fractional coordinates	Site symmetry
Li	3a	3	(0, 0, 0)	$-3m$
Co	3b	3	(0, 0, 1/2)	$-3m$
O	6c	6	(0, 0, $z \approx 0.24$)	3m

[Here, each row corresponds to one crystallographic orbit.]

2. Silicon (Diamond cubic structure)

Crystal system: Cubic

Space group: Fd-3m (No. 227)

Atom	Wyckoff position	Multiplicity	Fractional coordinates	Site symmetry
Si	8a	8	(1/8, 1/8, 1/8)	$-43m$

3. CoFe_2O_4 (Inverse spinel ferrite)

Crystal system: Cubic

Space group: Fd-3m (No. 227)

Atom	Wyckoff position	Multiplicity	Fractional coordinates	Site symmetry
Fe^{3+}	8a (tetrahedral)	8	(1/8, 1/8, 1/8)	$-43m$
$\text{Co}^{2+} / \text{Fe}^{3+}$	16d (octahedral)	16	(1/2, 1/2, 1/2)	$-3m$
O^{2-}	32e	32	($x \approx 0.26$, x, x)	m

Crystal directions and planes:

In a crystal there exist directions and planes which contain a large number of atoms.

For understanding crystallography, the concept of directions and planes play an important role. It is necessary to locate their directions and planes for crystal analysis.

Crystal directions:

To understand the crystal directions let us first take the case of two dimensions. The crystal direction is described by writing the first integer point (x, y) in square brackets [] through which the directional line passes. For example, in figure the direction OA, OB, OC, OD are written as [1 3], [1 1], [3 2], [4 1] respectively.

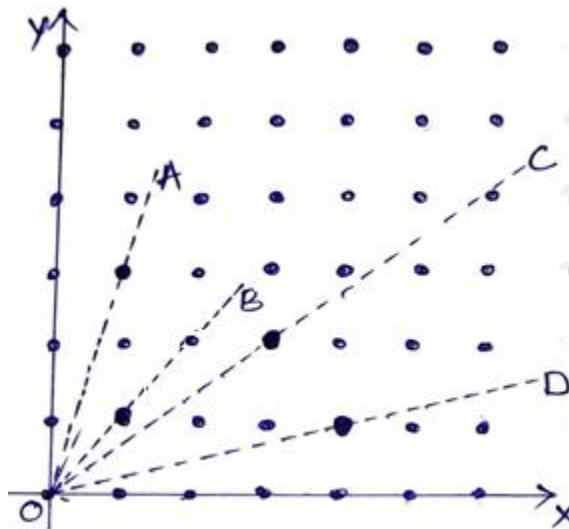


Fig. Crystal Direction

In similar way the crystal direction in three dimensions is written as [x y z] where (x, y, z) is the coordinate of the first whole number point through they pass. The square brackets can be used to indicate the direction (note: commas are unnecessary). In general, the indices of a direction are written as [u v w].

A negative index is indicated by a bar over the digit.

For example, the positive X-axis [1 0 0], and for negative X-axis $[\bar{1} 0 0]$.

- ❖ Directions [3 3 3] and [2 2 2] directions are identical to [1 1 1]. In such a case lowest combination of integers i.e. [1 1 1] is used to specify the directions.

Crystallographic Planes:

The crystal lattices may be considered as made up of an aggregate of set of equidistant parallel planes passing through lattice points which are known as lattice planes. For a given lattice, the lattice planes can be chosen in different

ways as shown in the figure below. These planes are known as crystallographic planes or atomic planes.

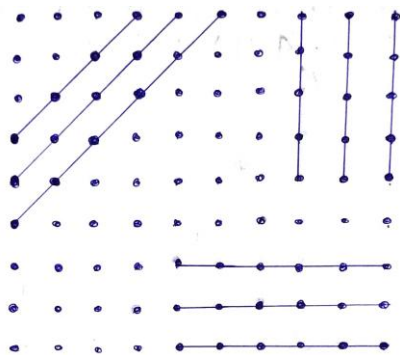


Fig. Crystal Planes

Miller Indices:

Miller developed a method to designate a set of parallel planes in a crystal by three numbers (h k l) which are known as miller indices.

Definition: "Miller indices are the reciprocals of the intercepts made by a plane on the three crystallographic axes when reduced to smallest integers".

Procedure for finding miller indices:

1. Find out the intercepts of the desired plane on the three coordinate axes /crystallographic axes (x, y, z).
2. Express these intercepts in the terms of axial lengths / lattice parameters (a, b, c) [Ex: 1a, 3b, c/2].
3. Take the reciprocals of the above intercepts. [1/1, 1/3, 2/1]
4. Reduce these numbers /reciprocal intercepts into smallest three integers which have the same ratio (this can be done by taking LCM of numerical values from the above reciprocals and multiplying with LCM to get smallest numbers) [Ex: LCM of 1, 3, 1 will be 3, then $\rightarrow \left[\frac{1}{1} \times 3 : \frac{1}{3} \times 3 : \frac{2}{1} \times 3\right] = 3 : 1 : 6$.
5. Write the result in the parenthesis as (h k l) which are known as miller indices of the crystal plane i.e. (3 1 6).

Important features of miller indices:

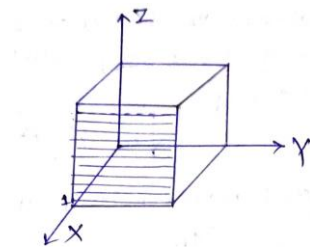
1. All the equidistant parallel planes have the same miller indices (h k l).
2. When a plane is parallel to one of the three crystallographic axes, then the intercept made by that plane with that axes is 'infinity'.
3. Miller indices are defined for a set of parallel planes but not for individual planes.

4. If any plane making the intercept on negative axis then the corresponding miller indices is represented by a bar on the top of miller index. Ex: $(\bar{2} 0 0)$.
5. A plane passing through the origin is defined in terms of a parallel planes having non zero intercepts.
6. For a cubic system, the distance 'd' between the adjacent planes of a set of parallel planes of the indices (h k l) is given as, $d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$.

Significance/ importance of Miller Indices:

- ✓ Miller indices are used to specify directions and planes in crystals. It is necessary to locate their directions and planes for crystal analysis.
- ✓ In Materials Science it is important to have a notation system for atomic planes since these planes influence,
- ✓ Optical properties
- ✓ Reactivity
- ✓ Surface tension
- ✓ Dislocations.

Example 1:



1. The plane cuts X- axis at '1a' distance, Y-axis at ' ∞ ' and Z-axis at ' ∞ '.
2. Expressing these intercepts in terms of lattice parameters (1a, ∞ b, ∞ C) and the intercepts are (1, ∞ , ∞).
3. Reciprocals of the above intercepts $(\frac{1}{1}, \frac{1}{\infty}, \frac{1}{\infty})$ i.e. (1, 0, 0).
4. Reduce the reciprocals of intercepts into smallest integers it is noted that here the reciprocals of the intercepts already smallest whole numbers. Hence, the smallest numbers are (1, 0, 0).
5. Write these smallest numbers in parenthesis i.e. (1 0 0) and these are the miller indices.

1. Different crystal planes with miller indices:

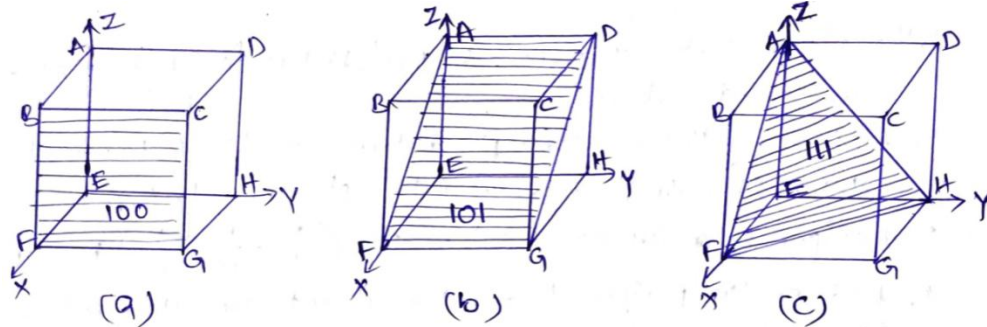
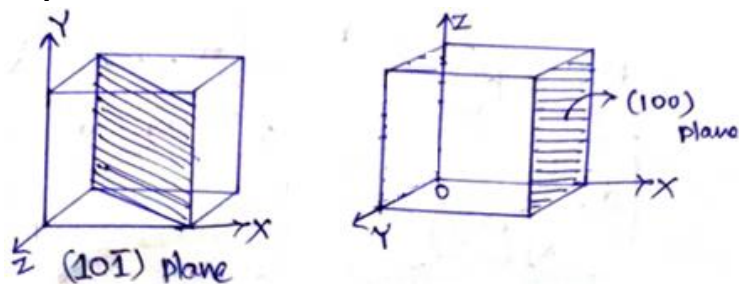


Fig. (a) BCGF plane with miller indices (100) **(b)** ADGF plane with miller indices (101), and **(c)** AHF plane with miller indices (111).

2. Different crystal planes with miller indices:



Examples: Draw the planes corresponding to following miller indices of (112) & (120).

1. (112):

- Given miller indices $(h\ k\ l) = (1\ 1\ 2)$
- Reciprocals of miller indices $= (\frac{1}{1}, \frac{1}{1}, \frac{1}{2})$
- Take the reciprocals as intercepts i.e. $(1, 1, \frac{1}{2})$
- Identify $(1, 1, \frac{1}{2})$ on the crystallographic axes and joining them to draw (112) plane.

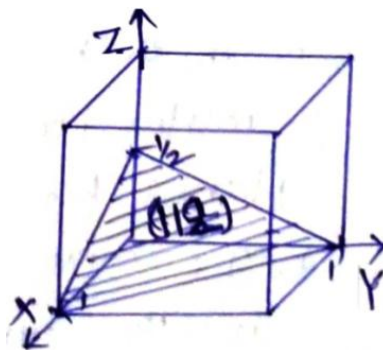


Fig: (112) plane with intercepts of $(1, 1, \frac{1}{2})$ on coordinate axes.

2. (120):

- Given miller indices $(h\ k\ l) = (1\ 2\ 0)$
- Reciprocals of miller indices $= (\frac{1}{1}, \frac{1}{2}, \frac{0}{1}) = (1, \frac{1}{2}, \infty)$

- Now the intercepts are $(1, \frac{1}{2}, \infty)$
- The above intercepts are marked on three crystallographic axes and joining them then the (120) plane is formed/drawn.

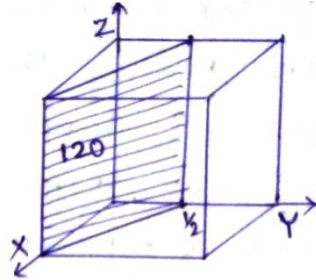


Fig: (120) plane with intercepts of $(1, \frac{1}{2}, \infty)$ on coordinate axes.

Miller Directions (or) Direction of a line:

The direction of a line in space can be defined with the coordinate through which it is passing. In order to indicate the direction, usually the square brackets [] are used. For a cubic system, the direction of a line is defined as a normal to the plane whose miller indices are (h k l).

Miller direction is represented by [h k l].

Example: Draw the direction [120] of a simple cubic crystal.

[120] direction:

- Along X and Y axes 1 & 2 units are moved and the position 'p' is marked.
- Since the directional unit corresponding to Z- axis is '0'.
- OP is joined, it gives [120] direction.

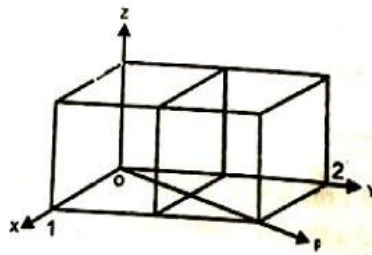


Fig: [120] direction.

[112] direction:

- Along X, Y and Z axes 1, 1 & 2 units are moved and the position 'p' is marked.
- OP is joined, it gives [112] direction.

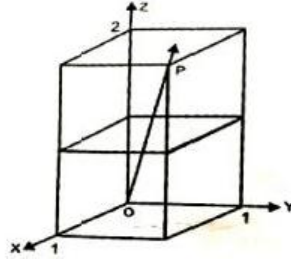


Fig: [112] direction.

Inter-planar spacing:

The separation between successive lattice planes is known as inter-planar spacing. It is defined as spacing between a plane (h k l) and other parallel plane passing through the origin. The inter-planar spacing is denoted by d_{hkl} . The inter-planar distance for cube, tetragonal and orthorhombic crystals for which $\alpha=\beta=\gamma=90^\circ$ can be calculated by the following formula, $d_{hkl} = \frac{1}{\sqrt{(\frac{h}{a})^2 + (\frac{k}{b})^2 + (\frac{l}{c})^2}}$

Derivation: Let a plane ABC making the intercepts X-axis, Y-axis and Z-axis at $\frac{a}{h}$, $\frac{b}{k}$, $\frac{c}{l}$ respectively and another plane is passing through the origin. The normal distance between the two planes is called inter-planar spacing i.e. ON and $ON = d$. ON is a normal to the plane ABC and making an angle with X, Y, Z-axis, i.e. α, β, γ .

$$\text{From the figure, } \cos\alpha = \frac{ON}{OA} = \frac{d}{a/h} \quad \text{----- (1)}$$

$$\cos\beta = \frac{ON}{OB} = \frac{d}{b/k} \quad \text{----- (2)}$$

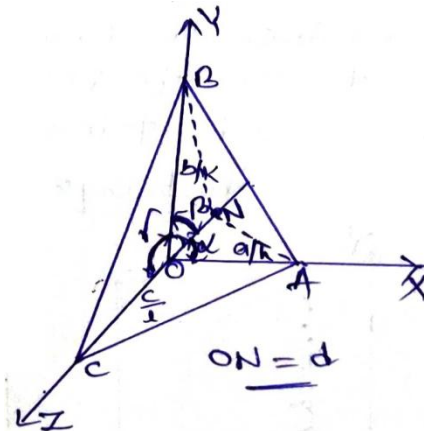
$$\cos\gamma = \frac{ON}{OC} = \frac{d}{c/l} \quad \text{----- (3)}$$

According to the law of direction cosines

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$$

$$\left(\frac{hd}{a}\right)^2 + \left(\frac{kd}{b}\right)^2 + \left(\frac{ld}{c}\right)^2 = 1$$

$$d_{hkl} = \frac{1}{\sqrt{(\frac{h}{a})^2 + (\frac{k}{b})^2 + (\frac{l}{c})^2}}$$



Special Cases:

(i) **Cubic system:** for Cubic system, $a = b = c$ then, $d_{hkl} = \frac{1}{\sqrt{(\frac{h}{a})^2 + (\frac{k}{a})^2 + (\frac{l}{a})^2}}$

$$\text{Therefore, } d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

For the hexagonal lattice ($a = b, c, \alpha = \beta = 90^\circ, \gamma = 120^\circ$), ex. Graphite

$$d(hkl) = \left(\frac{4}{3} \frac{h^2 + k^2 + hk}{a^2} + \frac{l^2}{c^2} \right)^{-\frac{1}{2}}$$

For the tetragonal lattice ($a = b, c, \alpha = \beta = \gamma = 90^\circ$)

$$d(hkl) = \left(\frac{h^2 + k^2}{a^2} + \frac{l^2}{c^2} \right)^{-\frac{1}{2}}$$

For the orthorhombic lattice ($a, b, c, \alpha = \beta = \gamma = 90^\circ$)

$$d(hkl) = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right)^{-\frac{1}{2}}$$

For the monoclinic lattice ($a, b, c, \alpha = \gamma = 90^\circ, \beta > 90^\circ$)

$$d(hkl) = \left(\frac{h^2}{a^2 \sin^2 \beta} + \frac{k^2}{b^2} + \frac{l^2}{c^2 \sin^2 \beta} - \frac{2hl \cos \beta}{ac \sin^2 \beta} \right)^{-\frac{1}{2}}$$

For the triclinic lattice ($a, b, c, \alpha, \beta, \gamma$), the most general case,

$$d(hkl) = \left(1 - \cos^2 \alpha - \cos^2 \beta - \cos^2 \gamma + 2 \cos \alpha \cdot \cos \beta \cdot \cos \gamma \right)^{\frac{1}{2}} \left(\frac{h^2}{a^2} \sin^2 \alpha + \frac{k^2}{b^2} \sin^2 \beta + \frac{l^2}{c^2} \sin^2 \gamma + \frac{2kl}{bc} (c \cdot \cos^2 \beta - \cos \alpha) + \frac{2lh}{ca} (c \cdot \cos^2 \gamma - \cos \beta) + \frac{2hk}{ab} (c \cdot \cos^2 \alpha - \cos \gamma) \right)^{-\frac{1}{2}}$$

Reciprocal Lattice:

Due to different orientations of crystallographic planes, it becomes difficult to visualize all such planes at a time.

This problem was simplified by P.P. Ewald by developing a new type of lattice known as Reciprocal Lattice.

The reciprocal lattice is a Fourier-transformed, wave-vector representation of a real crystal lattice, where each reciprocal vector points perpendicular to specific lattice planes and its magnitude relates inversely to their spacing.

Reciprocal lattice is crucial for understanding diffraction (like X-ray) and electron behavior in solids.

If the real-space primitive vectors are **a**, **b**, **c**, the reciprocal vectors are:

$$\mathbf{a}^* = \frac{2\pi(\mathbf{b} \times \mathbf{c})}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}; \mathbf{b}^* = \frac{2\pi(\mathbf{c} \times \mathbf{a})}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}; \mathbf{c}^* = \frac{2\pi(\mathbf{a} \times \mathbf{b})}{\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})}$$

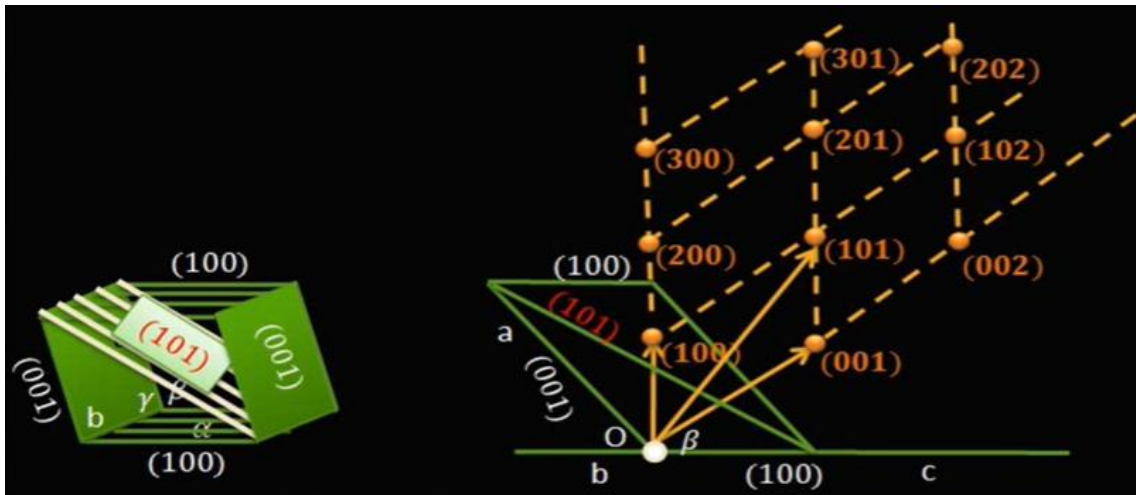
Any reciprocal lattice vector is:

$$\mathbf{G} = h\mathbf{a}^* + k\mathbf{b}^* + l\mathbf{c}^*$$

Construction of Reciprocal Lattice:

- Take any arbitrary point in the direct lattice as origin.
- Draw the normals to all set of planes present in the direct lattice from the origin.
- The length of the each normal should be the reciprocal of the interplanar spacing of the particular set of planes.
- Then place the lattice point at the end points of normals. The collection of such points will form a lattice which is called as Reciprocal Lattice.

The following figure represents a monoclinic lattice and it's reciprocal lattice.



Key properties:

1. Direct lattice is the reciprocal lattice to its own reciprocal lattice.
2. Volume of the unit cell in reciprocal lattice is inversely proportional to the volume of unit cell in direct lattice.
3. Each point in the reciprocal lattice correspond to a particular set of parallel planes in the direct lattice.
4. Reciprocal lattice of simple cubic (SC) lattice is simple cubic (SC).
5. But body centered cubic (BCC) and face centered cubic (FCC) are reciprocal to each other.

Diffraction of X-rays by crystal:

Introduction:

Diffraction effects can be observed only if the spacing between the lines ruled on the grating is of the order of magnitude of the wave length of light used. In case of diffraction of light, the spacing between the lines of grating element is in the order of 10^{-6} m.

X-rays are discovered by Rontgen. X-rays are produced when a beam of fast-moving electrons strikes a solid target material (tungsten or molybdenum). The wavelengths of X-rays ranging from $0.01\text{\AA}$ to $100\text{\AA}$ and are not deflected by electric field and magnetic fields. X-rays are the electromagnetic radiations with a wavelength of the order of a $\text{\AA}$ (Angstrom), i.e. $1/1000^{\text{th}}$ of wavelength of visible light. Therefore, to observe diffraction pattern of X-rays the grating should have a spacing $1/1000^{\text{th}}$ of the grating used for the diffraction of light.

Laue suggested that the crystal consists of arrays of atoms in three dimensions and it can be used for the diffraction of X-rays because the spacing between the atoms of a crystal is of the order of $1\text{\AA}$. The diffraction centers in the crystal are not in one plane thus the crystal act as three-dimensional grating.

Bragg's Law:

Crystals serve as grating to the X-rays to produce the diffraction of X-rays. W.H Bragg and W.L. Bragg while studying the properties of X-rays studied the X-ray diffraction pattern of few crystals. W.L. Bragg first derived a relation between interplanar spacing and angle of reflection.

"Bragg's law states that the X-rays are reflected from different parallel planes of a crystal interfere constructively when the path difference is the integral multiple of the wavelength of X-rays"

According to W.H Bragg and W.L. Bragg, the diffraction is due to the reflection of some incident X-rays from the various set of parallel crystal planes (known as Bragg's law) which contain a large number of atoms. When the two consecutive planes scatter waves in phase with each other, then the path difference must be equal to integral multiple of the incident wavelength.

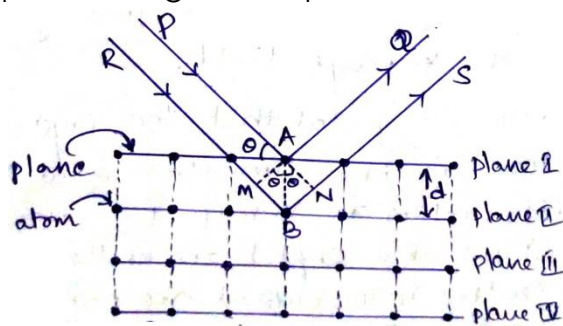


Fig. Reflection of X-rays from the crystal planes

Let us consider a set of parallel lattice planes of a crystal separated by a distance 'd' apart. A narrow beam of X-rays of wavelength ' λ ' be incident upon these planes at angle ' θ '. The beam will be reflected in all directions by the atoms of various atomic planes. The condition for the reflected waves to be all in phase is that path difference between two reflected waves must be an integral multiple of wavelength.

Consider two parallel rays PA and RB of the incident beam which are scattered by two atoms A and B in adjacent layers as shown in Fig. Constructive interference takes place only between those scattered rays that are parallel and whose path differ exactly $\lambda, 2\lambda, 3\lambda$ and so on., i.e. the path difference Δ must be $n\lambda$, where 'n' is an integer. Draw AM and AN perpendicular to the rays RB and BS. Path difference between the X-rays along PQ and RS are given by

$$\Delta = MB + BN = AB \sin\theta + AB \sin\theta$$

$$d \sin\theta + d \sin\theta$$

$$\Delta = 2d \sin\theta \text{ ----- (1)}$$

Where $AB = d$ is the interplanar spacing. i.e. spacing between the atomic planes. For constructive interference to take place, the path difference

$$\Delta = n\lambda \text{ ----- (2)}$$

Where $n = 1, 2, 3$ etc. for first order, second order, third order maxima etc.

From equation (1) & (2) **$\rightarrow 2d \sin\theta = n\lambda$**

This relation is known as Bragg's law for X-ray diffraction.

Different directions in which intense reflections will be produced can be obtained by giving different values to θ i.e., For 1st maxima $\sin\theta_1 = \lambda/2d$

For 2nd maxima $\sin\theta_2 = 2\lambda/2d$

For 3rd maxima $\sin\theta_3 = 3\lambda/2d$.

The intensities of maxima's go on decreases with increase in order of the spectrum.

Generally, the Bragg scattering is regarded as Bragg reflections and hence these set of parallel planes are known as Bragg planes. At certain glancing angles, reflections from these planes are in phase with each other and hence they reinforce each other to produce maximum intensity. For other angles, the reflections from different planes are out of phase and hence reinforce to produce either Zero or extremely feeble intensity.

Since the maximum possible value for $\sin\theta$ is 1 then **$n\lambda/2d \leq 1$ (or) $n\lambda \leq 2d$**

This sets limitation on wavelength. The wavelength λ should not exceed twice the inter-planar spacing for diffraction to occur.

Applications of X-ray diffraction:

X-ray diffraction procedures will reveal the details about,

- ✓ Arrangement of atoms
- ✓ Dimensions of Unit cell
- ✓ Grain size or crystallite size
- ✓ Orientations of poly-crystalline materials
- ✓ Imperfections or disorders in the unit cell
- ✓ Type of unit cell or crystal structure etc.

X-ray diffraction methods:

There are three main X-ray diffraction methods by which the crystal structure can be analyzed.

1. The Laue method (for single crystals)
2. The Powder method (for finely divided crystals or polycrystalline specimen powders)
3. The rotating crystal method (for single crystals)

Powder Diffraction method:

Powder X-ray diffraction method is usually carried for polycrystalline materials. This method reveals the information regarding the size of the crystal, presence of imperfections, distortion, and preferred orientations of crystals etc. The given polycrystalline material is grinded to get fine powders and this powder can be taken either in a capillary tube made up of non-diffracted material fixed at the centre of cylindrical camera known as Debye Scherrer camera whose length is small compared to diameter. A collimated beam of monochromatic X rays is produced by passing the X rays through a filter and collimator. Then the monochromatic X rays enter the camera through a small hole and strike the powder specimen. Since the specimen contains large number of small crystallites with random orientations. The diffraction takes place for those values of 'd' and 'θ' satisfy the Bragg's condition i.e. $2d\sin\theta = n\lambda$.

If θ is the glancing angle corresponding to a value of 'd' for one set of planes the reflected ray will be inclined at an angle '2θ'. The open of the cone is the point of contact of X-ray with the specimen.

Different cones are observed for different sets of 'd' and 'θ' for a particular value of 'n' and also for different combinations of 'θ' and 'n' for a particular values of 'd'. The transmitted ray move out of the camera through an exit hole located diametrically opposite to the entrance hole. A part of each reflected cone is recorded on the film which is positioned around the camera. The recorded lines are in the form of arc's as shown in figure.

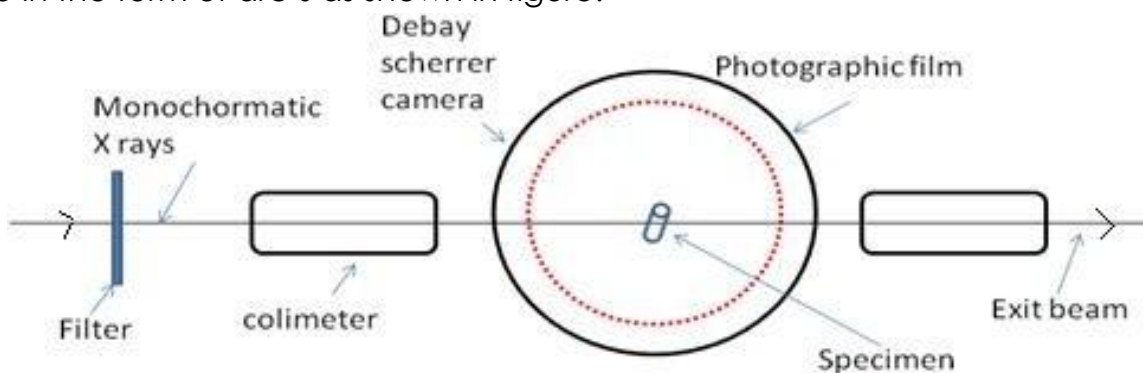


Fig. (a)

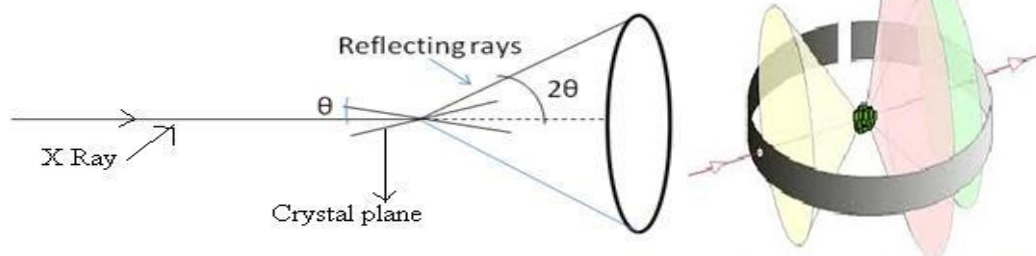


Fig. (b)

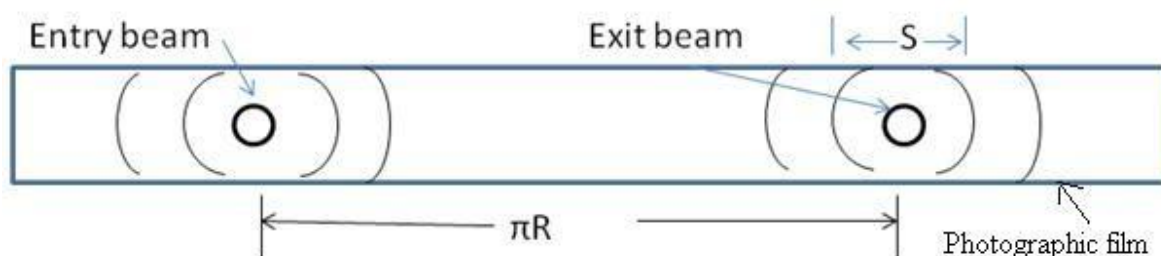


Fig. (c)

Fig. Powder method **(a)** Experimental arrangement **(b)** Reflection of X ray from the crystal plane, and **(c)** Reflection ray pattern on the photographic film.

The film is removed from the camera, processed and flattened. It shows the diffraction lines and the holes for the incident and transmitted beams. The distance between two successive arcs "S" is measured and using the relation, $4\theta = \frac{S}{R} \left(\frac{180}{\pi} \right)$, Where 'R' is the radius of the camera. A list of θ values can be obtained. Since the wavelength λ is known, substituting these θ values in Bragg's formula, a list of inter-planar spacing 'd' can be calculated. Each spacing is the distance between neighboring planes (hkl). From the ratio of inter-planar spacing, the type of the lattice can be identified as well as lattice parameters are evaluated.

❖ **Advantages of powder X-ray diffraction method:**

- ✓ The powder technique has the advantage that it does not require large single crystals and almost any substance can be ground into powder.
- ✓ The experiment is easy to perform
- ✓ Small crystals of any substance can be easily obtained.

❖ **Determination of crystal structure:**

Let us consider the 1st order spectrum from three planes one after the other reflecting a given monochromatic X-ray beam of wavelength λ at glancing angles $\theta_1, \theta_2, \theta_3$.

According to Bragg's law, $2d\sin\theta = n\lambda$ and let $n=1$ then, $2d\sin\theta = \lambda \rightarrow \frac{1}{d} = \frac{2\sin\theta}{\lambda}$

Now calculate 'd' values for the given planes (hkl) as shown below

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = \sin\theta_1 : \sin\theta_2 : \sin\theta_3$$

The ratio of $\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}}$ for different cubic lattices are given below:

Simple cubic $1 : \sqrt{2} : \sqrt{3}$

Body centered cubic $1 : \frac{1}{\sqrt{2}} : \sqrt{3}$

Face centered cubic $1 : \sqrt{2} : \frac{\sqrt{3}}{2}$

For example, in case of NaCl crystal first order reflection maxima occurs at 5.9° , 8.4° , 5.2° for (100), (110), (111) planes respectively. So,

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = \sin 5.9^\circ : \sin 8.4^\circ : \sin 5.2^\circ$$

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = 0.1028 : 0.1461 : 0.0906 \rightarrow 1.00 : 1.421 : 0.881$$

$$\frac{1}{d_{100}} : \frac{1}{d_{110}} : \frac{1}{d_{111}} = 1 : \sqrt{2} : \frac{\sqrt{3}}{2}$$

Hence NaCl has a face centered cubic structure.

By knowing the values of glancing angles ratio, the ratio of Inter planer spacing and hence the type of the lattice can be identified.

Crystal Defects

Introduction:

Crystal is defined as regular and periodic arrangement of atoms in three dimensions. But no crystal is perfectly regular which means crystal possesses some defects.

The disturbance occurred in the regular arrangement of atoms or molecules (may be at a point / along a line / in a region) is called crystal defect or imperfection.

The concept of ideal crystal is nothing but a crystal without any defect or imperfection. But in the real crystals this is not possible, because in every crystal at least few imperfections exist. Some defects are important and most useful whereas other defects are useless.

Classification of crystal defects:

Crystal defects or imperfections can be classified on the basis of their geometry and are four types i.e.

1. Point defects (Zero dimensional)
2. Line defects (One dimensional)
3. Surface defects (Two dimensional)
4. Volume defects (Three dimensional)

(1) Point defects:

Point defects / imperfections are also called as Zero-dimensional imperfections. In a crystal lattice, point defect is one which is completely local in its effect, i.e. a vacant lattice site. The point defects are formed due to the missing of atoms or due to the presence of extra / interstitial atom in the crystal.

The presence of point defects in a crystal increases its internal energy compared to that of the perfect crystal. Further they produce distortions inside the crystal structure up to few atomic diameters. Due to this there is no bonding at distorted point. Hence the mechanical strength at this point gets reduced. The point defects change the electrical resistance of a crystal. Point defects are created during the crystal growth.

The point defects are again classified into four types they are:

- (a) Vacancy,
- (b) Interstitials,
- (c) Compositional and
- (d) electronic defects.

(a) Vacancy defect:

When an atom is missing from its normal position, then the defect is called as vacancy defect. (Or) An empty site of an atom in a crystal is called vacancy. Vacancies may occur as a result of imperfect packing during the original crystallization or they may arise from thermal vibrations of atoms at higher temperatures because as thermal energy is increased then there is a higher probability that individual atoms will jump out from their original position of lowest energy.

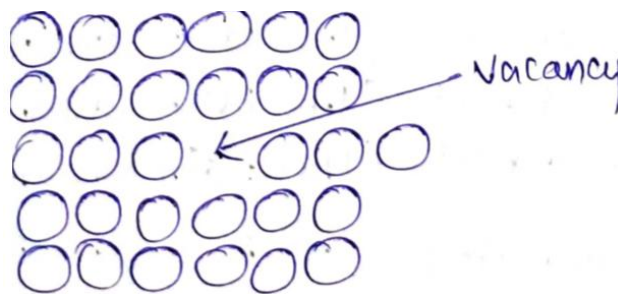


Fig. Vacancy defect.

Schottky Defect:

The schottky defect is closely related to vacancies. It is formed when a cation and anion are removed from a normal lattice site and replaced in an average position on the surface of the crystal.

A pair of one cation and one anion can be missing from an ionic crystal. Hence

ion vacancies are called as schottky defect.

The concentration of schottky defect decreases the density of the crystal. Crystals such as NaCl, KCl, KBr etc. show Schottky defect.

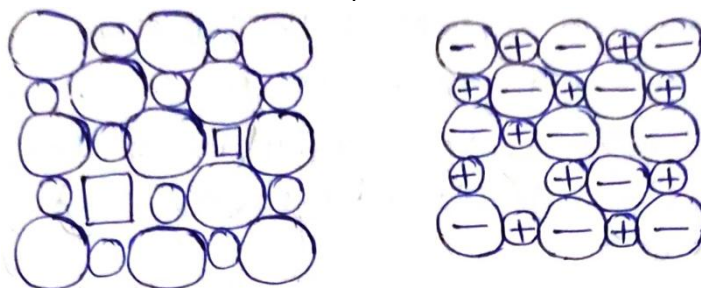


Fig. Schottky defect.

(b) Interstitial defect:

This type of point defect arises when an atom occupies a definite position in the lattice that is not normally occupied in the perfect crystal. (Or) This defect occurs when an additional atom occupies empty space between the atoms. Due to this additional atom some mechanical stress is present on the surrounding atoms. The additional atom which occupies the empty space is called interstitial atom. The amount stress depends on the size of the interstitial atom. This interstitial atom tends to push the surrounding atoms and produces atomic distortion. This defect arises when the density of packing (packing fraction) is low.

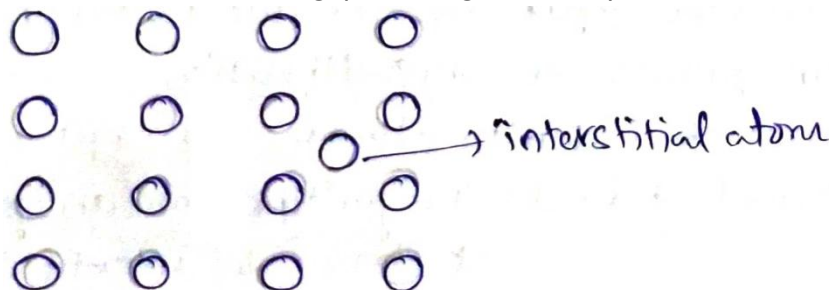


Fig. Interstitial defect.

Frenkel Defect:

The frenkel defect is closely related to the interstitial defect. In the case of ionic crystals, "an ion is displaced from the lattice into an interstitial site" is called as frenkel defect. As cations are generally the smaller ions, it is possible for them to get displaced into the void space present in the lattice.

Anions do not get displaced like cations as the void space is just too small for their size. A frenkel defect is the combination of one cation vacancy and one cation interstitial defect i.e. vacancy + interstitial = frenkel defect.

The concentration of frenkel defects does not change the density of the crystal and overall electrical neutrality of the crystal. Crystals such as CaF_2 , AgBr, AgI etc. show the frenkel defect.

✓ Frenkel and schottky defects together are called as "intrinsic defects".

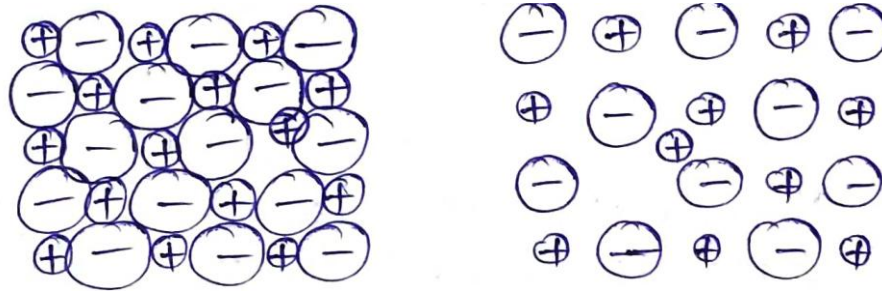


Fig. The ion moved from lattice position to interstitial: Frenkel defect.

(c) Compositional defects:

The compositional defects are formed due to presence of unwanted impurities or due to the doped impurities during the process of crystallization.

The presence of impurity atom in the crystal leads to the compositional defect.

Compositional defects play an important role in the semiconductors which are specially prepared for diodes and transistors etc.

The compositional defects are again classified into two types: (i) Substitutional impurity (ii) Interstitial impurity.

(i) Substitutional impurity:

A substitutional impurity refers to a foreign (metal) atom that replaces a parent/regular atom in the crystal lattice.

For example, in extrinsic semiconductor either third or fifth group atoms occupy the sites of Silicon or Germanium atoms. If the size of the substitutional impurity is same as the parent /regular atom then the amount of strain around will be less, otherwise it will be more.

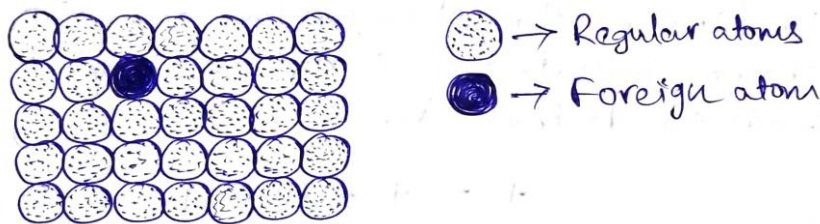


Fig. Substitutional impurity.

(ii) Interstitial impurity:

An interstitial impurity refers to a small sized foreign atom lodging the void space in the crystal without disturbing any of the parent atoms from their sites.

The small sized atoms (lower atomic number atoms) such as hydrogen may

fit into the interstitial spaces and these atoms are known as interstitial spaces and are responsible for the interstitial impurity.

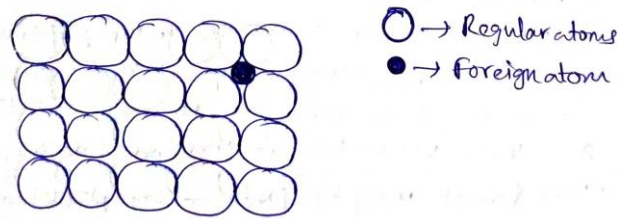


Fig. Interstitial impurity.

(d) Electronic defects:

Electronic defects occur as a result of errors in charge distribution in solids.

Non-uniformity of charge or charge distribution in the crystal is referred to as an electronic defect.

In the presence of impurity atoms (such as substitution and interstitial atoms) can vary the uniform distribution of electronic charge in the crystal, so, the presence of these defects leads to electronic defects.

In semiconductors, temperature variation changes the charge concentration, so the variation of temperature leads to electronic defects.

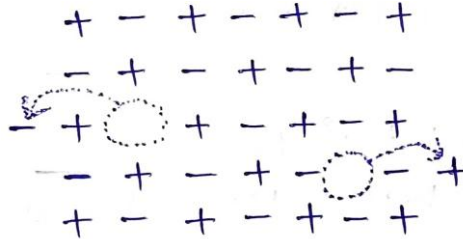


Fig. Electronic defect.

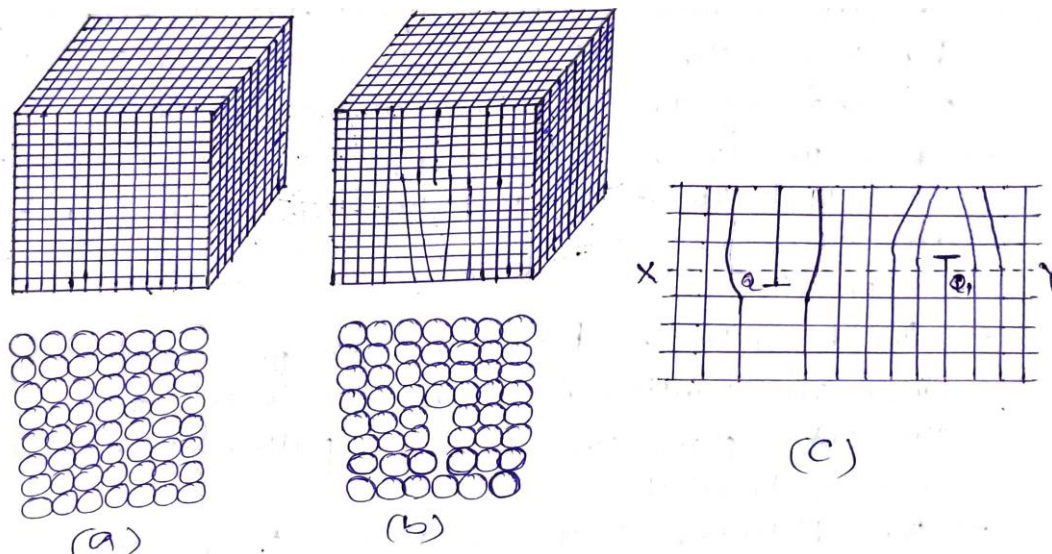
(2) Line defects:

- Line defects are known as one dimensional defects. These defects are also known as dislocations.
- These defects appear along a line in the crystal geometry.
- These dislocations occur generally due to the presence of incomplete atomic planes which arise due to growth accidents, thermal stress etc.
- The presence of dislocations in the crystal decreases its mechanical strength.
- A dislocation may be defined as a disturbed region between two substantially perfect parts of a crystal.
- It arises due to a part of atomic plane is displaced from its symmetrically stable positions in the array.
- In general, the line defects are again classified into two categories, (a) Edge dislocation and (b) Screw dislocation.

(a) Edge dislocation:

- An edge dislocation is created in the crystal by introducing an extra half plane (or any extra plane that does not extend up to the base of the crystal).
- Figure(a) shows a perfect crystal, the top sketch depicting a three-dimensional view and the bottom one showing the atoms on the front face.
- From the lower sketch of figure (b), it can be clearly seen that the atoms above the edge of the extra plane are squeezed together and are in a state of compression.
- The bond lengths have been compressed to smaller than the equilibrium value just below the edge of the extra plane the atoms are pulled apart and are in a state of tension.
- This distorted configuration extends along the edge of the crystal. There is an extra energy due to the distortion in the region immediately surrounding the edge of the incomplete plane.
- If one of the plane of atoms glides over another separated by an integral multiple of interatomic distance is called slip, and slip plane is the plane in which slip has taken place. In figure (C), XY- is the slip plane.
- If the incomplete extra plane is above the slip plane, then the edge dislocation is positive and is represented by the symbol ' $\perp$ '.
- If the incomplete extra plane is below the slip plane, then the edge dislocation is negative and is represented by the symbol ' $\top$ ' as shown in figure (c).

Definition of Edge Dislocation: If one of the vertical planes does not extend to full length but ends in between within the crystal, then it is called as edge dislocation.



(b) Screw dislocation:

Screw dislocation is defined as a displacement of the atoms in one part of a crystal relative to the rest of the crystal, forming a spiral ramp around the dislocation line as shown in figure.

The plane ABCD is the slipped area, AD dislocation line. The name screw follows the new character of atomic planes which are transferred such that as one move around the dislocation line in circuit. Speed of movement of a screw dislocation is less than the edge dislocation.

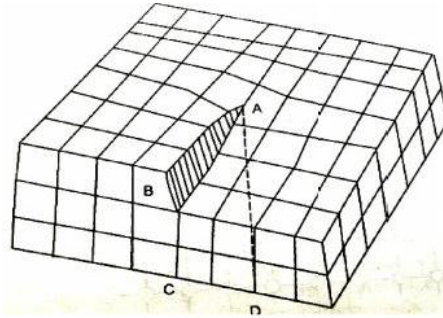


Fig. Screw dislocation

❖ **Burger's vector:**

- Dutch physicist, Jan Burger introduced a vector, which represents the direction and magnitude of dislocations in crystals, is called Burger's Vector. It is represented by ' $\vec{b}$ '.
- Burgers vector indicates how much and in what direction the dislocation has been shifted. The magnitude and direction of the dislocation can be understood from the concept of Burger's vector.
- Burger's vector is proportional to the square root of the elastic energy of the dislocation.
- In most metallic materials, the magnitude of the Burgers vector is equal to the interatomic spacing of the material, since a single dislocation will offset the crystal lattice by one close-packed crystallographic spacing unit.
- In edge dislocations, the Burgers vector and dislocation line are at right angles to one another. In screw dislocations, they are parallel.

Burger's circuit:

Dislocation in a crystal is represented by a closed loop is called Burger's circuit.

❖ **Burger's vector ($\vec{b}$) in Edge dislocation:**

- As shown in figure (a), let consider a point 'P'. From this point, we go up atom by atom by 6 steps, then move towards right by 5 steps, and move down by 6 steps and finally move towards left by 5 steps to reach the starting point 'P'.

Thus the circuit is closed.

- In figure (b), atoms are missing along edge, called edge dislocation. In this, circuit is failed to close itself in the region containing edge containing dislocation. Then the vector is necessary to close the circuit is called Burger's vector. Now we have to move an extra step QP to return to P in order to close the circuit.
- Burger's vector $\vec{b} = \vec{QP}$.
- In Edge dislocation, Burger's Vector ($\vec{b}$) is Perpendicular to dislocation line.

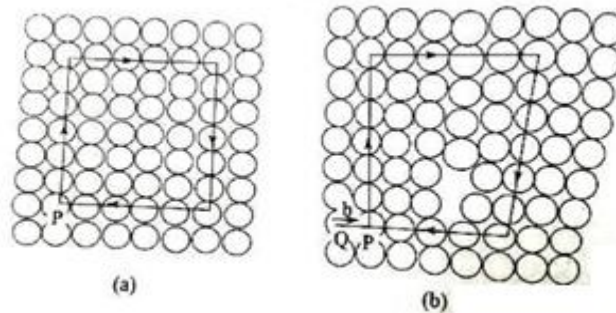


Fig. Burger's vector in edge dislocation.

❖ **Burger's vector ($\vec{b}$) in Screw dislocation:**

In figure, atoms are missing and gives dislocation in the form of screw called, screw dislocation. In this, circuit is failed to close itself in the region containing screw dislocation. Then the vector is necessary to close the circuit is called Burger's vector.

In Screw dislocation, Burger's Vector ($\vec{b}$) is Parallel to dislocation line.

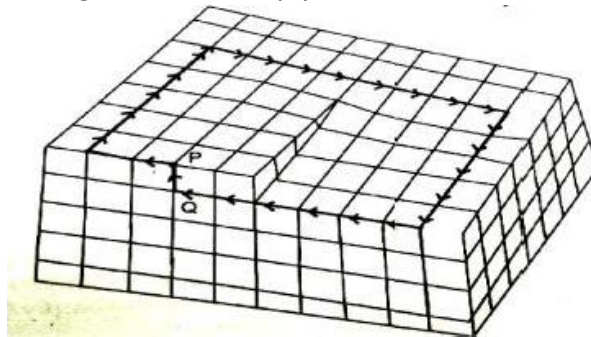


Fig. Burger's vector in Screw dislocation.

(3) Surface defects:

Surface defects/imperfections arise from a change in the stacking of atomic planes on or across a boundary.

In surface defects, the crystalline irregularity extends in two dimensions. Surface defects are also called as plane defects. The surface imperfections refer to regions of distortions that lie about a surface having a thickness of few atomic diameters.

The surface defects are classified into two types, (a) External surface imperfections, and (b) Internal surface imperfections.

- In external surface defects, the external surface of a crystal is an imperfection in itself as the atomic bonds do not extend beyond the surface.
- The internal surface defects arise from a change in the stacking of atomic planes across a boundary. The internal surface defects further classified into 4 types, (i) Grain boundaries, (ii) Tilt boundaries, (iii) Twin boundaries, and (iv) Staking faults.

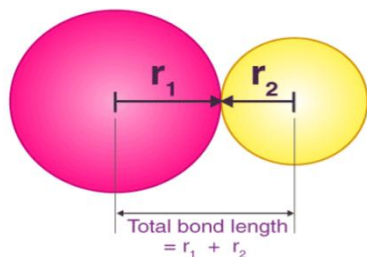
(4) Volume defects:

Volume defects such as cracks may arise in crystals when they are grown or while they are used.

- When clusters of atoms are missing, a large vacancy or void is created which are also a volume defect /imperfection.
- Foreign particle inclusions, large voids and non-crystalline regions which have the dimensions of the order of 1 or 3 nm are also called as volume defects/ imperfections.

Bond length and bond angles:

- Bond length is the distance between the nuclei of two covalently bonded atoms. It represents the equilibrium distance where the attractive and repulsive forces between the atoms are balanced. Bond length is influenced primarily by the bond order, which is the number of shared electron pairs between two atoms.
- It is typically measured in picometers (pm) or angstroms ($\text{\AA}$).
- Larger atoms form longer bonds due to the increased distance between their nuclei.
- Bond length affects molecular geometry and physical properties like boiling and melting points.
- Bond angles are the angles formed between adjacent bonds at the central atom in a molecule.



- A common observation is that as bond angles increase (atoms pushed closer), bond lengths decrease, and vice versa, as longer bonds reduce repulsive forces between electron clouds.

Brillioun Zones (or) E-K Diagram:-

Brillioun zones are the boundaries that are marked by the values of wave vector K , in which the electrons can have allowed energy values. These represent the allowed values of K of the electrons in 1D, 2D & 3D.

We have, the energy of the electron in a constant potential box is,

$$E = \frac{n^2 h^2}{8ma^2} \quad \text{----- (1), Where } a = \text{length of the box.}$$

$$\text{But, } K = \frac{n\pi}{a} \Rightarrow K^2 = \frac{n^2 \pi^2}{a^2}$$

$$\frac{n^2}{a^2} = \frac{K^2}{\pi^2} \quad \text{----- (2) Substitute eqn. (2) in (1), we get}$$

$$E = \frac{K^2 h^2}{8m\pi^2} \quad \text{i.e., } E \propto K^2 ; \text{ It represents parabolic equation.}$$

Graph:-

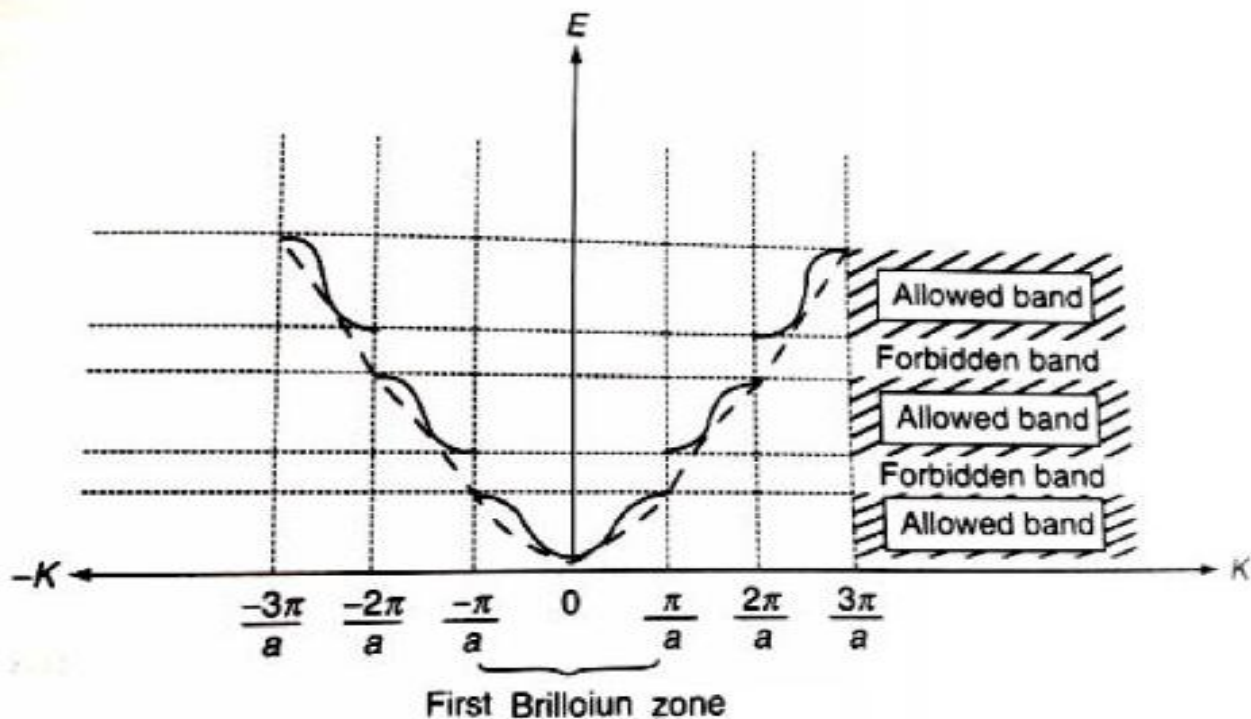
A graph is drawn between the total energy (E) and the wave vector K , for various values of K . i.e., $K = \frac{n\pi}{a}$; $n = \pm 1, \pm 2, \pm 3, \dots$

$\Rightarrow$ 1st Brillouin Zone (1st BZ): $-\frac{\pi}{a}$ to $+\frac{\pi}{a}$;

$\Rightarrow$ 2nd Brillouin Zone (2BZ): $-\frac{\pi}{a}$ to $-\frac{2\pi}{a}$ and $+\frac{\pi}{a}$ to $+\frac{2\pi}{a}$;

Similarly 3rd, 4th Brillouin Zones can be obtained.

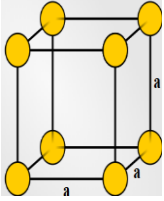
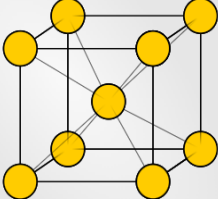
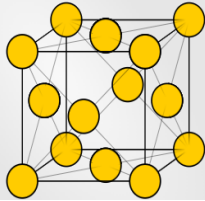
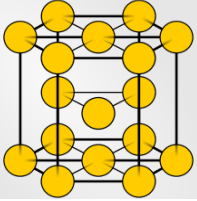
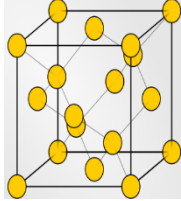
Brillioun Zones are separated by forbidden gap.



Some crystal structures:

1. **Atomic radius (r):** It is defined as half the distance between nearest neighboring atoms in a crystal.
2. **Nearest neighbor distance (2r):** The distance between the centers of two neighboring atoms is called as nearest neighbor distance. If 'r' is the radius of the atom, then the nearest neighbor distance is '2r'.
3. **Coordination Number (N):** Coordination number is defined as number of equidistant nearest neighbors that an atom has in a given structure. More closely packed structure has greater N.
4. **Atomic packing factor:** Atomic packing factor is the ratio of volume occupied by atoms in a unit cell to the total volume of the unit cell.

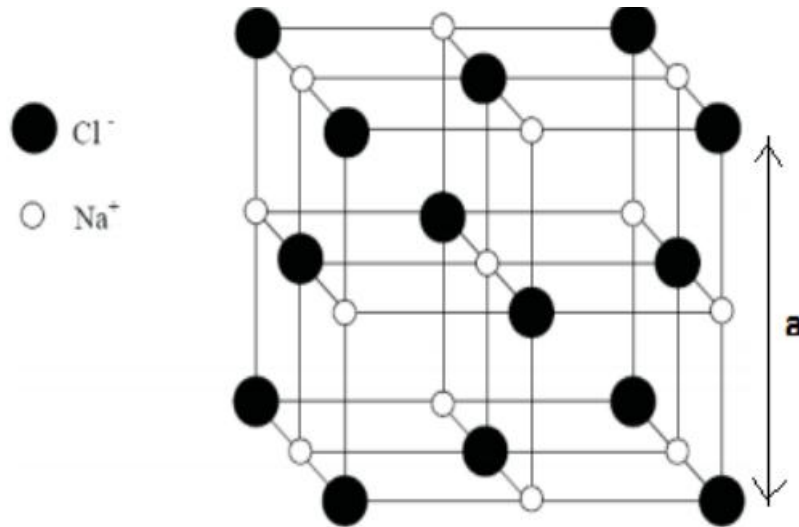
$$APF = (n \times V_a) / V_u$$
 where, $n \rightarrow$ effective number of atoms, $V_a \rightarrow$ Volume of atom, $V_u \rightarrow$ Volume of unit cell.
5. **Void space:** The void space in the unit cell is the vacant space left unutilized in the cell. It is often expressed in percentage, Void space = $(1 - APF) \times 100$. The void space is most commonly known as interstitial space.

Properties	Simple (Primitive) Cubic (SC)	Body Centered Cubic (BCC)	Face Centered Cubic (FCC)	Hexagonal Close Packed	Diamond
Unit Cell					
Volume of Unit cell	a^3	a^3	a^3	$\frac{3\sqrt{3}}{2}a^2c$	a^3
Number of atoms per unit cell	1	2	4	6	8
Coordination number	6	8	12	12	4
Nearest neighbor distance (2r)	a	$\frac{\sqrt{3}a}{2}$	$\frac{\sqrt{2}a}{2}$	a	$\frac{\sqrt{3}a}{4}$
Atomic radius (r)	$\frac{a}{2}$	$\frac{\sqrt{3}a}{4}$	$\frac{\sqrt{2}a}{4}$	$\frac{a}{2}$	$\frac{\sqrt{3}a}{8}$
Atomic packing factor (APF)	0.52	0.68	0.74	0.74	0.34
Void space	48 %	32 %	26 %	26 %	66 %
Examples	Polonium	Sodium, Lithium, Iron, Barium	Silver, Aluminum, Nickel	Cadmium, Beryllium	Carbon, Germanium, Silicon

❖ Sodium chloride structure:

NaCl is an ionic crystal and in this crystal the sodium and chlorine ions are situated side-by-side in the FCC lattice.

Rock salt (NaCl) structure type is AX, anion packing is fcc, here the cation number is 6, anion is 6. A basis consist one Na and one Cl. position of Cl and Na atoms are: Cl : 0, 0, 0; 1/2, 1/2, 0; 1/2, 0, 1/2; 0, 1/2, 1/2; Na : 1/2, 1/2, 1/2; 0,0, 1/2; 0; 0, 1/2, 1/2; 1/2,0,0. The coordinate number of each Na and Cl atom in this structure is 6. Other examples of NaCl (AX) type structure are KBr, KCl, MgO, AgBr.



❖ Cesium Chloride (CsCl) Structure:

Cesium chloride (CsCl) structure type is AX at anion packing is body centered cubic. Two primitive cells in a cubic unit cell. each unit cell has two molecules (basis) of CsCl. Positions of Cl ion is at (0, 0, 0) and Cs ion is at (1/2, 1/2, 1/2). The Cs is situated at body center and 8 Cl ions at the corner of unit cell. Similarly, if we extend the unit cell we can see a Cl ion is surrounded by 8 Cs ions. Thus, the coordinate number of CsCl is 8. The other examples of CsCl type structure are RbCl, LiHg etc.

